

# Airframe Technology Master

## Lecture 09. Corrosion Prevention and Erosion prevention

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Introduction

Corrosion theory

- Basic corrosion processes
- Electrochemical corrosion

Typology of corrosion

Materials Vs corrosion

Corrosion protection

- Introduction
- Means to avoid the corrosion
- Aircraft corrosion areas (environmental categories)
- Practical approach to corrosion protection

References

## What is corrosion?

There are many definitions, this is one of them:

*“Corrosion is the destructive attack of a metal by a chemical or electrochemical reaction with its environment”*

There are other kind of damages suffered by the metals, but with a physical origin, the erosion or wear, these phenomena are not corrosion, but they can appear together with the corrosion what enhances the corrosion process, these corrosion processes are called erosive corrosion, corrosive wear or corrosion by friction.

The corrosion is a natural process in which a metallic element transforms in a more stable element, this element is an oxide.

Non metallic materials, plastic or wood do not suffer corrosion, they can degrade, crack, break, but not corrode.

## **Corrosion consequences**

The corrosion produces the degradation of the affected part until its complete destruction. The corrosion leads to huge economical and production losses.

The most important consequence is that the people security is permanently under risk.

The economical impact of the costs derived from the corrosion is about the 4% of the del G.D.P (Gross Domestic Product) of the industrialized countries.

**Basic corrosion processes:**Dry corrosion:

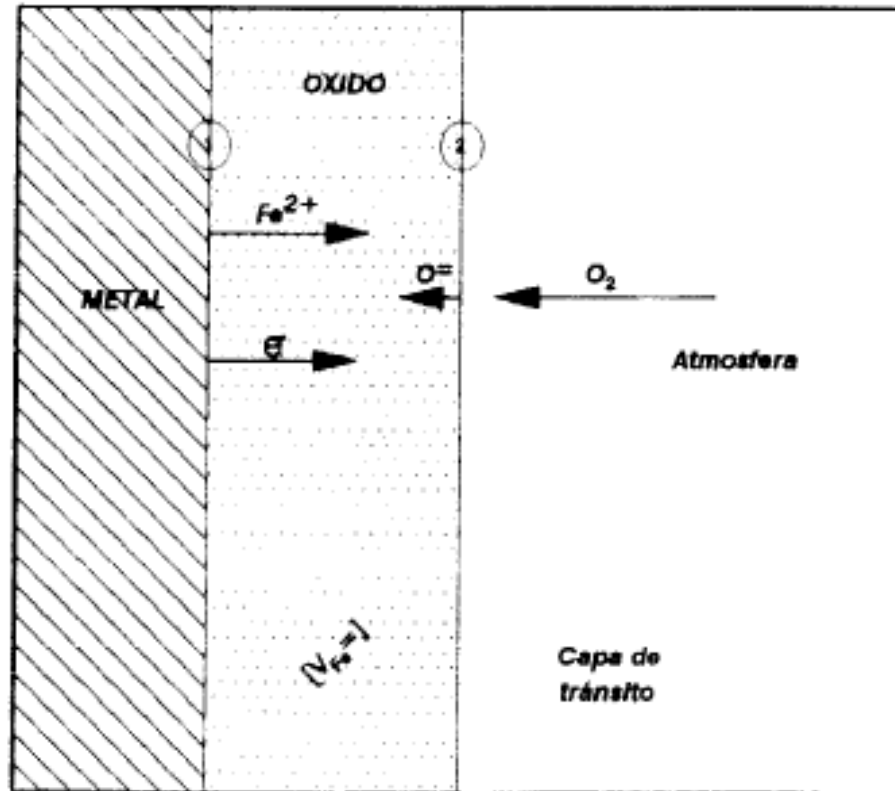
Is the corrosion that appears in absence of an electrolyte, is a chemical reaction.

The Cl, S and O ions present at the atmosphere enter the metal by diffusion, they react with the metal molecules generating metallic ions in the metal surface.

The high temperatures increase both the diffusion and the corrosion processes.

## Basic corrosion processes:

### Dry corrosion:



## Basic corrosion processes:

### Wet corrosion:

Appears in presence of an electrolyte, which is basically an electricity conduction liquid.

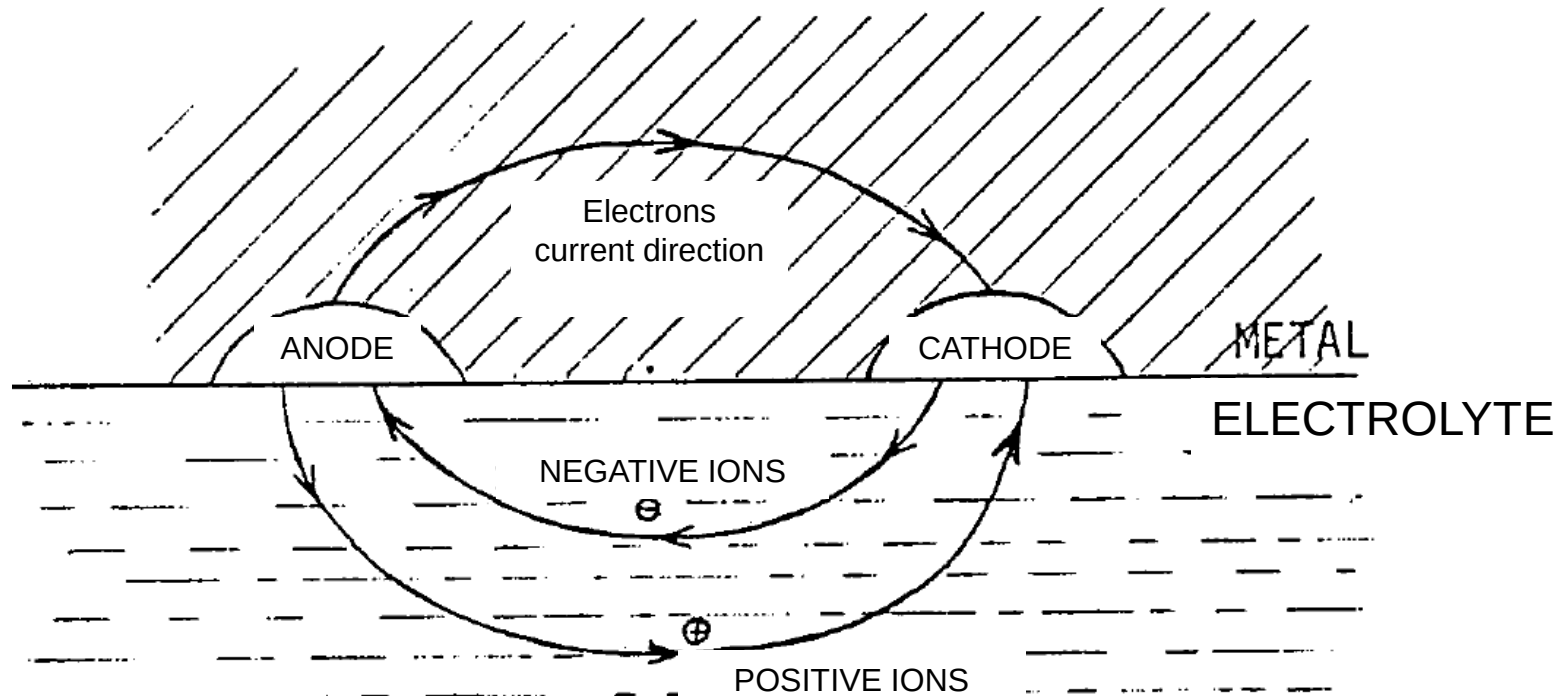
An electrical potential difference between two metals or between two zones of the same metal causes an electrical current through the electrolyte.

The electrical field help the ions transport, that's why it's called Electrochemical corrosion.

At not so high temperatures, this process is predominant over the dry corrosion, therefore this is the main corrosion process that affect to the metals life, and that's why most bibliography, examples and protection means are focused in this type of corrosion (wet corrosion or electrochemical corrosion).

## Basic corrosion processes:

### Wet corrosion (electrochemical)





## Electrochemical corrosion:

Three main actors are:

- Anode: Material with tendency to give up electrons, thus a oxidation (corrosion) reaction is produced (Anodic Reaction)
- Cathode: Material with tendency to get electrons, thus an reduction reaction is produced (Cathodic Reaction)
- Electrolyte: Is a liquid solution electrical conductive which is the corrosive mean.

## Electrochemical corrosion:

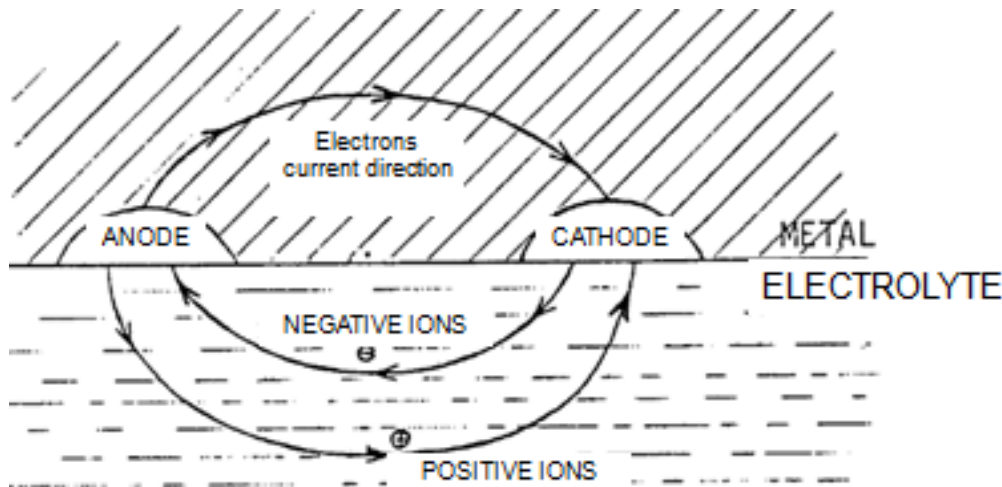


Fig.1. Electrodes belonging to the same part

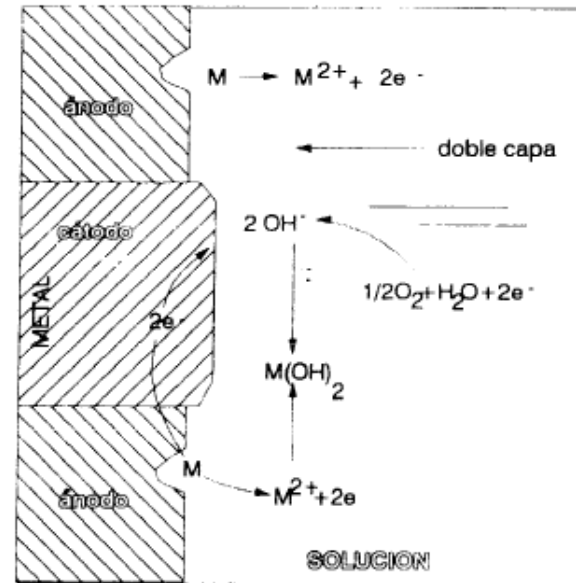


Fig.2. Electrodes from different parts

- Anodic Reaction:  $METAL \rightleftharpoons METAL^{+n} + n e^{-}$   
(oxidation = release e<sup>-</sup>)
  - Cathodic Reaction:  $2H^{+} + 2e^{-} \rightleftharpoons H_2$  (Acid environ.) + H<sub>2</sub>O  
(reduction = catch e<sup>-</sup>)  $\frac{1}{2} O_2 + H_2O + 2e^{-} \rightleftharpoons 2OH^{-}$  (Alkaline env.)
- Diagram illustrating the flow of electrons from the anodic reaction to the cathodic reaction, leading to the formation of corrosion products:  $MO_n$  and  $M(OH)_n$ .

**Electrochemical corrosion:**

The anode and the cathode can be two different metals in contact or two zones of the same part, this is possible due to several reasons: electrolyte with not uniform concentration in different areas, potential differences between the base metal and the alloys, between the grains and the grain limits, etc

The anode metal oxidizes generating positive ions which solve in the electrolyte and electrons which are transferred through the metal up to the cathode.

At the cathode the electrons coming from the anode contribute to the cathodic reaction in combination with Oxygen and water molecules in alkaline dissolutions or  $H^+$  ions in acid dissolutions.

The molecules resulting from the cathodic reaction react with the metallic positive ions from the anodic reaction generating metal oxides (acid environment) or hydroxides (alkaline environment).

**Electrochemical corrosion:**Electrode potential:

When a metal is immersed in an aqueous medium, the metal atoms tend to ionize giving electrons, ions are dissolved in solution and the electrons remain in the metal, this creates a potential difference between the metal and the dissolution.

To compare the potentials of various metals is usually taken as reference the hydrogen potential, which is assigned a value "0".

The potential of the oxidation reaction of an electrode has the same absolute value but sign contrary to the potential of the inverse reduction reaction, by convention, the electrode potentials sign is the sign of the reduction reaction.

Electrochemical corrosion:

Electrode potential:

Galvanic cell potential is:

$$E^0 = E^0_{\text{CÁTODO}} - E^0_{\text{ÁNODO}}$$

Example Fe<sup>3+</sup> (Cathode) / Al (Anode)

$$E^0 = 0,77 - (-1,68) = 2,45 \text{ V}$$

$E^0 > 0$  Spontaneous reaction. Generates an electrical field. Galvanic cell

$E^0 < 0$  Non spontaneous. Required an electrical field. Electrolytic cell.

ANODIC

CATHODIC

| REDUCTION REACTION                                                                                                                | $E^0$ (V)   |
|-----------------------------------------------------------------------------------------------------------------------------------|-------------|
| $\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Mg}(\text{s})$                                                         | -2.38       |
| $\text{Al}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Al}(\text{s})$                                                         | -1.68       |
| $\text{Mn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Mn}(\text{s})$                                                         | -1.18       |
| $\text{V}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{V}(\text{s})$                                                           | -1.13       |
| $\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$                                                         | -0.76       |
| $\text{Cr}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Cr}(\text{s})$                                                         | -0.74       |
| $\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Fe}(\text{s})$                                                         | -0.44       |
| $\text{Cr}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Cr}^{2+}(\text{aq})$                                                    | -0.42       |
| $\text{Cd}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cd}(\text{s})$                                                         | -0.40       |
| $\text{V}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{V}^{2+}(\text{aq})$                                                      | -0.26       |
| $\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ni}(\text{s})$                                                         | -0.25       |
| <b><math>2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})</math></b>                                         | <b>0.00</b> |
| $\text{Cu}^{2+}(\text{aq}) + \text{e}^- \rightarrow \text{Cu}^+(\text{aq})$                                                       | +0.16       |
| $\text{Cu}^+(\text{aq}) + \text{e}^- \rightarrow \text{Cu}(\text{s})$                                                             | +0.52       |
| $\text{CO}(\text{g}) + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{C}(\text{s}) + \text{H}_2\text{O}$                             | +0.52       |
| $[\text{AuI}_4]^- (\text{aq}) + 3\text{e}^- \rightarrow \text{Au}(\text{s}) + 4\text{I}^- (\text{aq})$                            | +0.56       |
| $\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{aq})$                                                    | +0.77       |
| $\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s})$                                                             | +0.80       |
| $[\text{AuBr}_4]^- (\text{aq}) + 3\text{e}^- \rightarrow \text{Au}(\text{s}) + 4\text{Br}^- (\text{aq})$                          | +0.85       |
| $[\text{AuCl}_2]^- (\text{aq}) + \text{e}^- \rightarrow \text{Au}(\text{s}) + 2\text{Cl}^- (\text{aq})$                           | +1.15       |
| $\text{Ag}_2\text{O}(\text{s}) + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{Ag}(\text{s})$                                      | +1.17       |
| $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}$ | +1.36       |
| $\text{Au}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Au}(\text{s})$                                                         | +1.52       |
| $\text{NiO}_2(\text{s}) + 4\text{H}^+ + 2\text{e}^- \rightarrow \text{Ni}^{2+}(\text{aq})$                                        | +1.59       |
| $\text{Au}^+(\text{aq}) + \text{e}^- \rightarrow \text{Au}(\text{aq})$                                                            | +1.68       |
| $\text{AgO}(\text{s}) + 2\text{H}^+ + \text{e}^- \rightarrow \text{Ag}^+(\text{aq}) + \text{H}_2\text{O}$                         | +1.77       |
| $\text{Au}^+(\text{aq}) + \text{e}^- \rightarrow \text{Au}(\text{s})$                                                             | +1.83       |
| $\text{Ag}^{2+}(\text{aq}) + \text{e}^- \rightarrow \text{Ag}^+(\text{aq})$                                                       | +1.98       |
| $\text{F}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{F}^-(\text{aq})$                                                           | +2.87       |

## **Electrochemical corrosion:**

### Electrochemical series:

Is a table with the metal arranged according to its electrode potential.

With the electrochemical potential values we know if the contact between two metals will produce corrosion of the most anodic metal (electronegative).

## Electrochemical corrosion:

### Electrochemical series:

Electrochemical series for metals commonly used in aeronautics:

| Metal or Alloy               | Potential, volts 0.1 N calomel scale |
|------------------------------|--------------------------------------|
| Magnesium                    | -1.73                                |
| Zinc                         | -1.10                                |
| Al. clad 7075                | -0.96                                |
| Al. clad 2014, Al. clad 2024 | -0.83                                |
| Cadmium                      | -0.82                                |
| 7075-T6                      | -0.81                                |
| 2024-T81                     | -0.80                                |
| 2014-T6                      | -0.78                                |
| 2024-T3 and T4               | -0.68                                |
| Mild steel                   | -0.58                                |
| Lead                         | -0.55                                |
| Tin                          | -0.49                                |
| Copper                       | -0.20                                |
| Stainless steel (300 series) | -0.09                                |
| Silver                       | -0.08                                |
| Nickel                       | -0.07                                |
| Chromium                     | - 0.49 to +0.018                     |

## **Electrochemical corrosion:**

### Corrosion degrees:

Taking into account the electrochemical potential we can classify the degree of corrosion for different couples:

A: Compatible couple.

B: The anodic metal could be slightly corroded by the cathodic metal

C: The anodic metal is corroded by the cathodic metal if there is an electrolyte

D: Couple not compatible except if adequate protection is provided



| Metal in contact                                 | Gold, Platinum<br>Silver | Stainless steel<br><u>austenítico</u> Cr/Ni<br>18/8 | Titanium | <u>Monel Inconel</u><br>Alloy Ni/Mo | Nickel-copper<br>Zinc Bronze | Copper<br>Bronze<br>Silver to the<br>nickel | Nickel | Tin Lead | Stainless steel<br>Cr/Ni 18/2 | Steel | Cadmium | Chrome | Al<br>Al alloys | Magnesium<br>Magnesium<br>alloys |
|--------------------------------------------------|--------------------------|-----------------------------------------------------|----------|-------------------------------------|------------------------------|---------------------------------------------|--------|----------|-------------------------------|-------|---------|--------|-----------------|----------------------------------|
| Considered metal                                 |                          |                                                     |          |                                     |                              |                                             |        |          |                               |       |         |        |                 |                                  |
| Gold Platinum<br>Silver                          | -                        | A                                                   | A        | A                                   | A                            | A                                           | A      | A        | A                             | A     | A       | A      | A               | A                                |
| Stainless steel<br><u>austenítico</u> Cr/Ni 18/8 | A                        | -                                                   | A        | A                                   | A                            | A                                           | A      | A        | A                             | A     | A       | A      | A               | A                                |
| Titanium                                         | A                        | A                                                   | -        | A                                   | A                            | A                                           | A      | A        | A                             | A     |         | A      | A               | A                                |
| <u>Monel, Inconel</u> , Alloy<br>Ni/Mo           | B                        | A                                                   | A        | -                                   | A                            | A                                           | A      | A        | A                             | A     | A       | A      | A               | A                                |
| Nickel-copper<br>Zinc Bronze                     | C                        | B/C                                                 | B/C      | B/C                                 | -                            | A                                           | A      | A        | B                             | A     | A       | B/C    | A               | A                                |
| Copper Bronze<br>Silver to the nickel            | C                        | B/C                                                 | B/C      | B/C                                 | B/C                          | -                                           | B/C    | B/C      | B/C                           | A     | A       | B/C    | A               | A                                |
| Nickel                                           | C                        | B/C                                                 | B/C      | B                                   | A                            | A                                           | -      | A        | B/C                           | A     | A       | B/C    | A               | A                                |
| Tin Lead                                         | C                        | B/C                                                 | B/C      | B/C                                 | B/C                          | B/C                                         | B      | -        | B/C                           | A/C   | A       | B/C    | A               | A                                |
| Stainless steel Cr/Ni<br>18/2                    | C                        | A                                                   | -        | A/C                                 | A/C                          | A/C                                         | A      | A        | -                             | A     | A       | A      | A               | A                                |
| Steel                                            | C                        | C                                                   | C        | C                                   | C                            | C                                           | C      | C        | C                             | -     | A       | C      | A               | A                                |
| Cadmium                                          | C                        | C                                                   | C        | C                                   | C                            | C                                           | C      | B        | C                             | C     | -       | C      | B               | A                                |
| Chrome                                           | A                        | A                                                   | A        | A                                   | A                            | A                                           | A      | A        | A                             | A     | A       | -      | A               | A                                |
| Al /Al alloys                                    | D                        | C                                                   | C        | C                                   | D                            | D                                           |        | C        | B/C                           | B/C   | A       | B/C    | -               | A                                |
| Magnesium<br>Magnesium alloys                    | D                        | C                                                   | C        | D                                   | D                            | D                                           | D      | C        | C                             | D     | B/C     | C      | B/C             | -                                |

| ABSCISSA \ ORDINATE         | Platinum | Gold/Carbon | Stainless steel | Titanium | AG - HG | Nickel | Copper alloy | Copper | Alu-bronze<br>Brass 30 % ZN | Silicon | Brass 50 % ZN | Bronze | Tin | Lead | NSA 3001 | Steels | Aluminium A5 | Cadmium | Chromium | Zinc | Manganese | Magnesium |
|-----------------------------|----------|-------------|-----------------|----------|---------|--------|--------------|--------|-----------------------------|---------|---------------|--------|-----|------|----------|--------|--------------|---------|----------|------|-----------|-----------|
| Platinum                    | 0        | 130         | 250             | 340      | 350     | 430    | 450          | 570    | 600                         | 685     | 700           | 770    | 800 | 840  | 940      | 1000   | 1090         | 1100    | 1200     | 1400 | 1470      | 1950      |
| Gold/Carbon                 | 130      | 0           | 110             | 210      | 220     | 300    | 320          | 440    | 470                         | 535     | 570           | 640    | 670 | 710  | 810      | 870    | 960          | 970     | 1070     | 1270 | 1340      | 1620      |
| Stainless steel             | 250      | 110         | 0               | 90       | 100     | 180    | 200          | 320    | 350                         | 415     | 450           | 520    | 550 | 590  | 690      | 750    | 840          | 850     | 950      | 1150 | 1220      | 1700      |
| Titanium                    | 340      | 210         | 90              | 0        | 10      | 80     | 110          | 230    | 280                         | 325     | 380           | 430    | 460 | 500  | 600      | 660    | 750          | 760     | 860      | 1050 | 1150      | 1610      |
| Silver - Mercury            | 350      | 220         | 100             | 10       | 0       | 80     | 100          | 220    | 250                         | 315     | 360           | 420    | 450 | 490  | 590      | 650    | 740          | 750     | 850      | 1050 | 1120      | 1600      |
| Nickel                      | 430      | 300         | 180             | 90       | 80      | 0      | 20           | 140    | 170                         | 235     | 270           | 340    | 370 | 410  | 510      | 570    | 650          | 670     | 770      | 970  | 1040      | 1520      |
| Copper alloy                | 450      | 320         | 200             | 110      | 100     | 20     | 0            | 120    | 150                         | 215     | 250           | 320    | 350 | 390  | 490      | 530    | 640          | 650     | 750      | 950  | 1020      | 1500      |
| Copper                      | 570      | 440         | 320             | 230      | 220     | 140    | 0            | 30     | 65                          | 95      | 130           | 200    | 230 | 270  | 370      | 430    | 520          | 530     | 630      | 830  | 900       | 1380      |
| Alu-bronze<br>Brass 30 % ZN | 600      | 470         | 350             | 260      | 170     | 30     | 0            | 0      | 65                          | 100     | 170           | 200    | 240 | 280  | 380      | 400    | 500          | 500     | 600      | 800  | 870       | 1350      |
| Silicon                     | 685      | 535         | 415             | 325      | 315     | 215    | 65           | 0      | 35                          | 105     | 135           | 175    | 200 | 240  | 340      | 335    | 425          | 435     | 535      | 735  | 805       | 1285      |
| Brass 50 % ZN               | 700      | 570         | 450             | 360      | 270     | 130    | 35           | 0      | 70                          | 100     | 140           | 170    | 200 | 240  | 340      | 300    | 390          | 400     | 500      | 700  | 770       | 1250      |
| Bronze                      | 770      | 640         | 520             | 430      | 340     | 200    | 105          | 70     | 0                           | 105     | 135           | 175    | 200 | 240  | 340      | 300    | 390          | 400     | 500      | 700  | 770       | 1250      |
| Tin                         | 800      | 670         | 550             | 460      | 370     | 230    | 135          | 30     | 0                           | 100     | 140           | 170    | 200 | 240  | 340      | 300    | 390          | 400     | 500      | 700  | 770       | 1250      |
| Lead                        | 840      | 710         | 590             | 490      | 410     | 270    | 175          | 70     | 0                           | 100     | 140           | 170    | 200 | 240  | 340      | 300    | 390          | 400     | 500      | 700  | 770       | 1250      |
| Light alloy<br>NSA 3001     | 940      | 810         | 690             | 590      | 510     | 370    | 240          | 175    | 70                          | 100     | 140           | 170    | 200 | 240  | 340      | 300    | 390          | 400     | 500      | 700  | 770       | 1250      |
| Steels                      | 1000     | 870         | 750             | 650      | 570     | 430    | 340          | 275    | 240                         | 335     | 300           | 320    | 350 | 380  | 480      | 500    | 600          | 600     | 700      | 900  | 970       | 1450      |
| Aluminium A5                | 1090     | 960         | 840             | 740      | 640     | 520    | 425          | 390    | 330                         | 320     | 320           | 320    | 350 | 380  | 480      | 500    | 600          | 600     | 700      | 900  | 970       | 1450      |
| Cadmium                     | 1100     | 970         | 850             | 760      | 670     | 530    | 435          | 400    | 330                         | 330     | 330           | 330    | 360 | 390  | 490      | 500    | 600          | 600     | 700      | 900  | 970       | 1450      |
| Chromium                    | 1200     | 1070        | 950             | 860      | 770     | 630    | 535          | 500    | 430                         | 430     | 430           | 430    | 460 | 490  | 590      | 600    | 700          | 700     | 800      | 1000 | 1070      | 1550      |
| Zinc                        | 1400     | 1270        | 1150            | 1050     | 970     | 830    | 735          | 700    | 630                         | 630     | 630           | 630    | 660 | 690  | 790      | 800    | 900          | 900     | 1000     | 1200 | 1270      | 1750      |
| Manganese                   | 1470     | 1340        | 1220            | 1120     | 1020    | 900    | 805          | 770    | 700                         | 700     | 700           | 700    | 730 | 760  | 860      | 870    | 970          | 970     | 1070     | 1270 | 1340      | 1820      |
| Magnesium                   | 1950     | 1620        | 1500            | 1400     | 1300    | 1180   | 1085         | 1000   | 910                         | 815     | 720           | 630    | 540 | 450  | 360      | 270    | 180          | 90      | 0        | 0    | 0         | 0         |

Below the line — the metal of the ordinate is attacked.  
Above the line — the metal of the abscissa is attacked.  
Between these 2 lines, the contact is practically neutral.

**Electrochemical corrosion:**Corrosion speed:

The corrosion speed is proportional to the galvanic cell intensity (speed of the electrons flow), the battery intensity depend on the anode and cathode potential and the electrical resistance of the electrolyte.

$$I \sim \frac{\Delta V}{R}$$

$\Delta V$  = Anode – Cathode potential difference

R: Circuit resistance which is the addition of the electric resistance plus the ionic resistance of the electrolyte.

## **Corrosion types:**

Uniform corrosion

Fretting corrosion

Pitting corrosion

Crevice corrosion

Cavitation

Erosion corrosion

Intergranular corrosion

Exfoliative corrosion

Stress corrosion

Fatigue corrosion

Galvanic corrosion

MIC corrosion

Filiform corrosion

Decarburation

## Uniform corrosion or surface corrosion

A uniform corrosion along the surface of a part is produced by contact with a corrosive environment also uniform, i.e., an atmosphere with moisture, and can be enhanced by high temperatures.

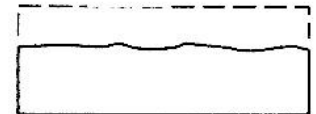
The material will undergo a constant decrease in the thickness, which can be predicted on the basis of the time. To avoid this type of corrosion surface protectors are applied taking into account the environment in which the aircraft operates, e.g. aluminium: TSA + primer + painting (area C). Steel: passivation.

In certain cases the plates or sheets of aluminum alloys are supplied with a cathodic protection to prevent corrosion before and during the production process until they receive the final surface protection, this protection is called cladding that consists of a thin layer of pure aluminum.

## Uniform corrosion or surface corrosion



Uniforme



## Fretting corrosion

This type of corrosion occurs when two metal surfaces in contact are highly loaded and subject to a slight vibrating or oscillating movement.

The friction destroys the protective film producing metal particles that are oxidized forming abrasive materials. This generates deep pitting, Galvanic pairs and cracks which accelerate the failure by fatigue of the parts.

Fretting corrosion can be divided into two cases:

- Relative displacement of parts by structural deformation.
- Parts in contact with relative movement.



## Fretting corrosion

The means to avoid the first case of fretting corrosion:

- Adequate fasteners pattern in order to distribute the load transfer reduce the micro strains.
- Use of nylon shims between the parts to be joined.
- For highly loaded holes: use bronze-niquel bushing and Hard Chromium CRES shafts (forbidden by REACH, see alternatives at slide 118).
- Use anti-friction paints, Teflon or sealant at matting surfaces.



## Fretting corrosion

The means to avoid the second case of fretting corrosion :

- Use friction resistant material couples alloys (see slides 26 to 28)
- Mating surfaces protection:
  - CRES: See slide 28
  - Titanium: Sulfuric Acid Anodizing.
  - Shot peening.
- Use of lubricant or Teflon coatings.

## Fretting corrosion

| Friction resistance of material couples | Steel | CRES | HSS  | Cold worked steel | Steel Casting | Silver Plating | Aluminum |
|-----------------------------------------|-------|------|------|-------------------|---------------|----------------|----------|
| Steel                                   | Low   | -    | -    | -                 | -             | -              | Low      |
| Aluminum                                | Low   | Low  | -    | -                 | -             | -              | Low      |
| Lead                                    | High  | -    | -    | -                 | High          | -              | -        |
| Zinc                                    | High  | -    | -    | -                 | -             | -              | -        |
| Silver Plating                          | High  | -    | -    | -                 | -             | -              | -        |
| Nylon                                   | High  | -    | -    | -                 | -             | -              | -        |
| HSS                                     | -     | Low  | High | -                 | -             | -              | -        |

## Fretting corrosion

| Friction resistance of material couples | Steel                           | CRES | HSS | Cold worked steel | Steel Casting                      | Silver Plating | Aluminum |
|-----------------------------------------|---------------------------------|------|-----|-------------------|------------------------------------|----------------|----------|
| Cold worked steel                       | -                               | -    | -   | High              | -                                  | -              | -        |
| Steel Casting                           | Molybdenum Disulfide (Molikote) | -    | -   | -                 | High (Molikote, Wolframium Sulfur) | Medium         | Low      |
| Silver Plating                          |                                 | -    | -   | -                 | -                                  | High           | Medium   |
| Niquel Plating                          | Low                             | -    | -   | -                 | -                                  | -              | Medium   |
| Copper Plating                          |                                 | -    | -   | -                 | Medium                             | -              | Medium   |
| Zinc                                    | Medium                          | -    | -   | -                 | Medium                             | -              | Medium   |
| Cadmium Plating                         | Medium                          | -    | -   | -                 | -                                  | -              | -        |

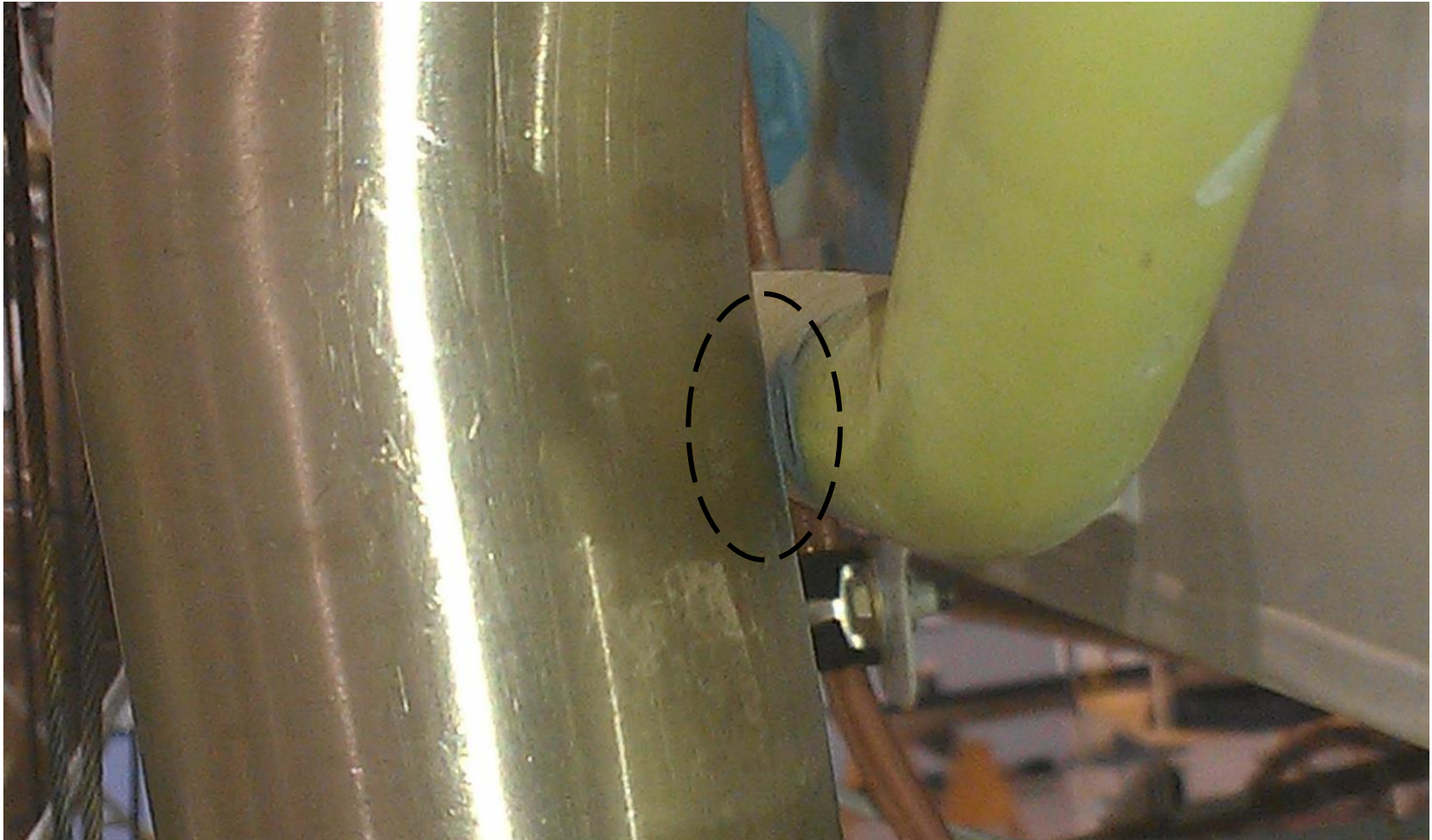
## Fretting corrosion

CRESS surfaces protection to avoid fretting:

| MATING MATERIAL       | CRESS SURFACE PROTECTION |                |                |
|-----------------------|--------------------------|----------------|----------------|
|                       | HARD CHROMIUM*           | NIQUEL PLATING | SILVER PLATING |
| CROMIUM               |                          | X              | X              |
| NIQUEL                | X                        |                | X              |
| PH STEEL              |                          | X              | X              |
| BRONCE AL-NI ALLOY    | X                        | X              | X              |
| COPPER BERILIUM ALLOY | X                        | X              | X              |

\*Chromium coating crack is avoided by previous niquel plating

## Fretting corrosion



## Pitting corrosion

It is a local corrosive attack, only small areas of the metal surfaces are attacked.

It is an electrochemical process, it is due to areas with different potentials in the metal surface and, therefore, some areas of the surface act as cathode and others as anode. It occurs as a process of local Anodic dissolution where metal loss is accelerated by the presence of a small anode and one cathode much greater.

Electrochemical potential differences appear due to:

- Dissimilar metals
- Protective coating discontinuities
- Mechanical stresses
- Pollutant particles at surface
- Oxygen concentration differences
- Environment pH differences
- Temperature differences

## Pitting corrosion

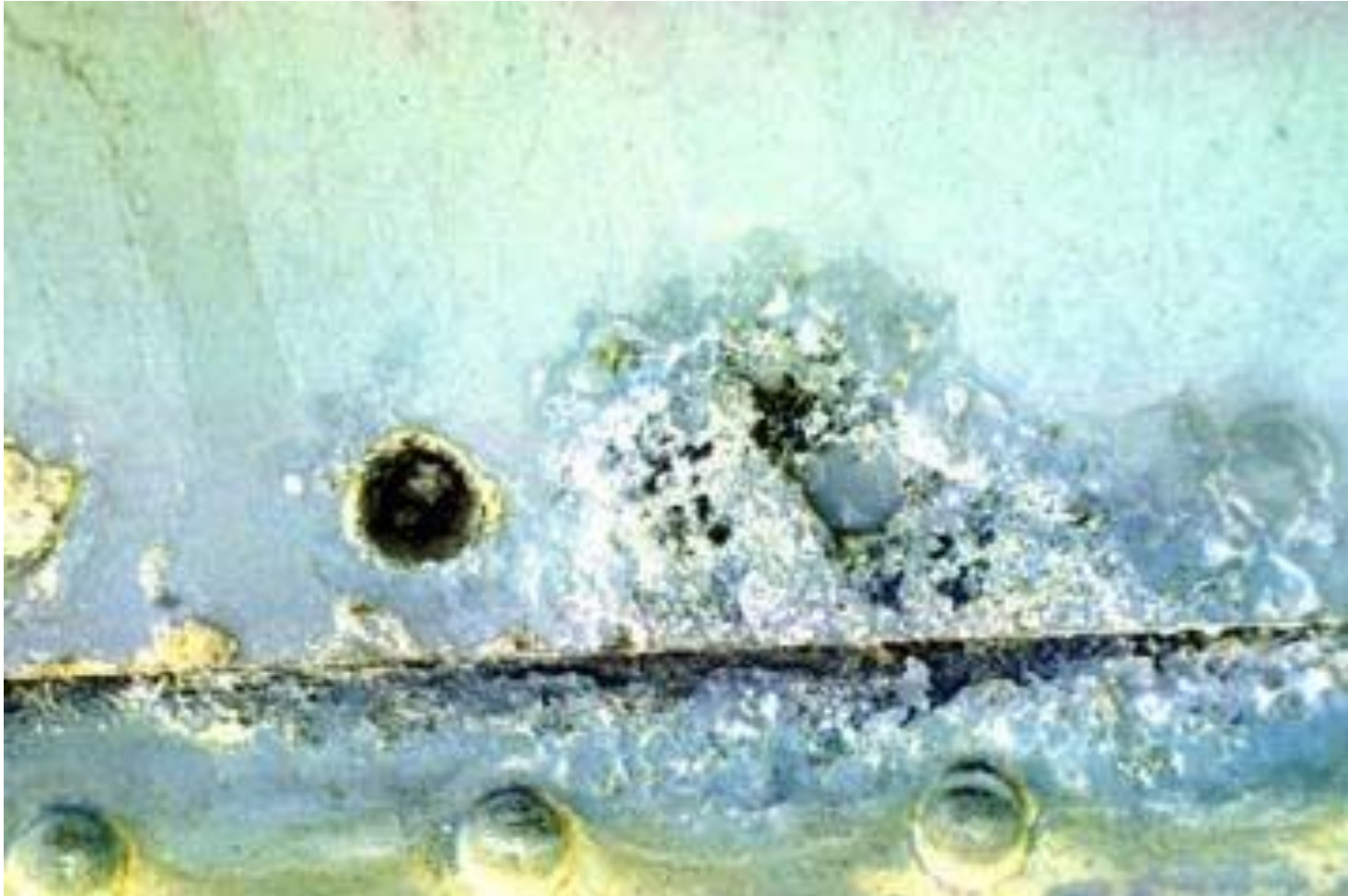
Most affected areas are riveted joints, welding, landing gear, lavatories, galleys, batteries, exhaust, specially A/C operating close to sea and industrial areas.

To avoid pitting corrosion:

- Avoid surface irregularities
- Keep surfaces clean
- Avoid protective coating damage
- Avoid galvanic couples
- Drainage
- Sealing



## Pitting corrosion





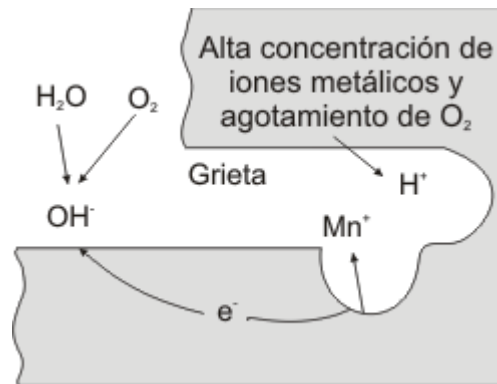
## Pitting corrosion



## Crevice corrosion

Occurs in small hollows formed at the joint between two parts or cavities in a piece.  
These cracks accumulate the dissolution that facilitates the corrosion of the piece.

This is called a stagnant anode corrosion, since that dissolution, unless it is removed, may never leave the fissure.



## **Crevice corrosion**

Way of avoid the crevice corrosion:

- Redesign to avoid crevice.
- Crevice sealing with non absorbent materials.

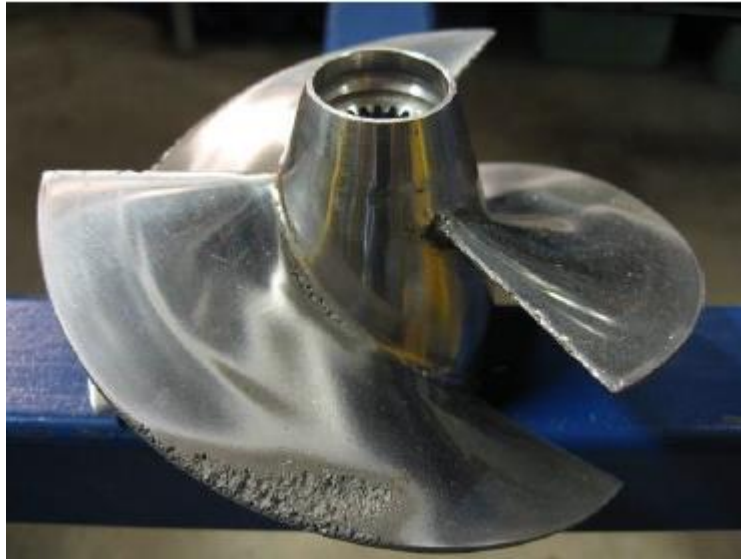
## **Cavitation corrosion**

It is caused by the formation and collapse of bubbles on the surface of a metal in contact with a liquid.

It is a similar phenomenon to that happens to the rear faces of the propellers of boats.

Generates a series of stings with honeycomb shape (local damage of surface protection of metal) that are sources of corrosion.

## Cavitation corrosion



## Erosion corrosion

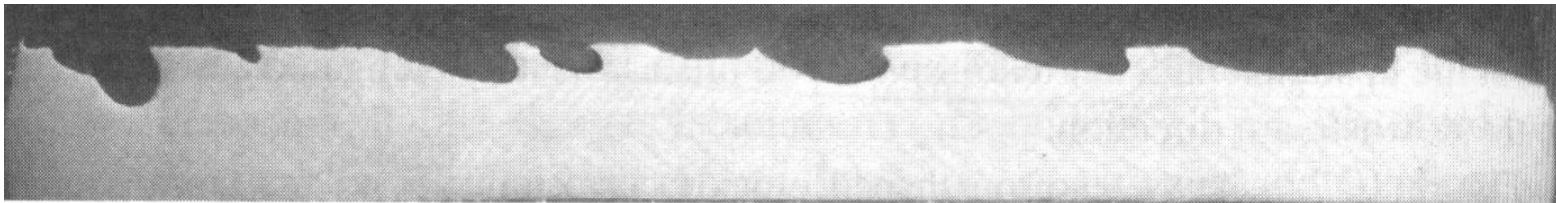
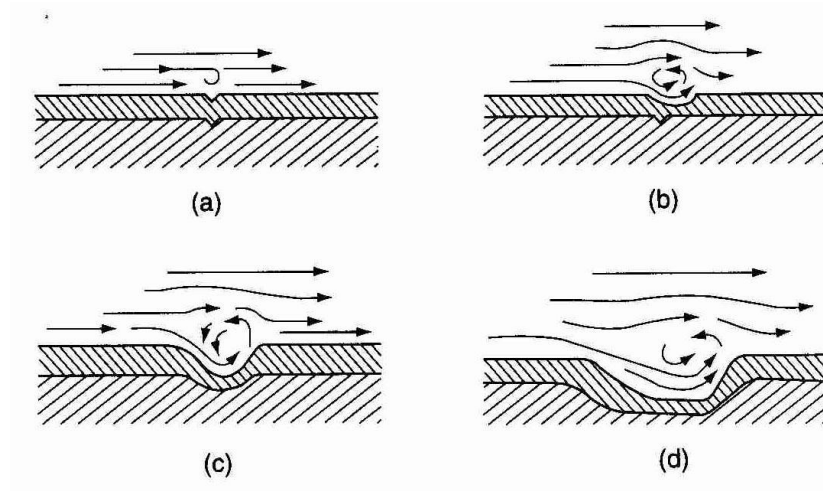
It can be defined as the combination of the effects of corrosion and mechanical wear.

When a metal is in the presence of a fluid with a high speed, the aggressiveness of corrosion increases since the fluid, liquid or gaseous, dissolves the metal ions that are on the surface of the workpiece. The greater are the speed and the number of dissolved particles the more virulent is this phenomenon.

This type of corrosion affects the parts exposed to the action of fluids in motion easy to be mechanically damaged. In desert environments, landing gears, propellers, leading edges, turbine parts, are severely attacked.

It is necessary to adopt wear resistant surface protectors.

## Erosion corrosion



## Intergranular corrosion

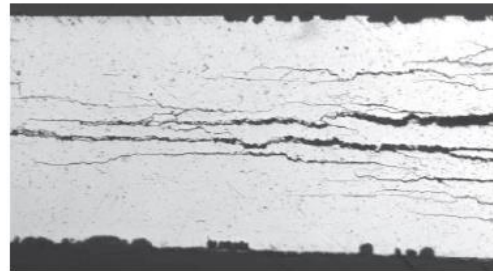
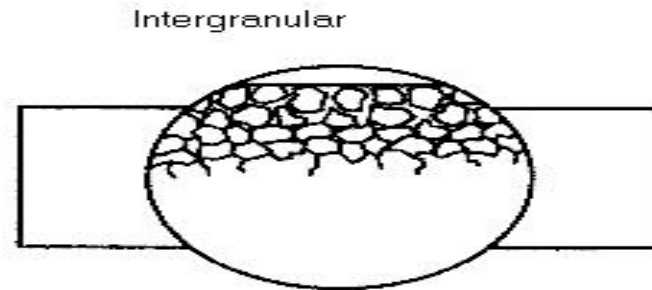
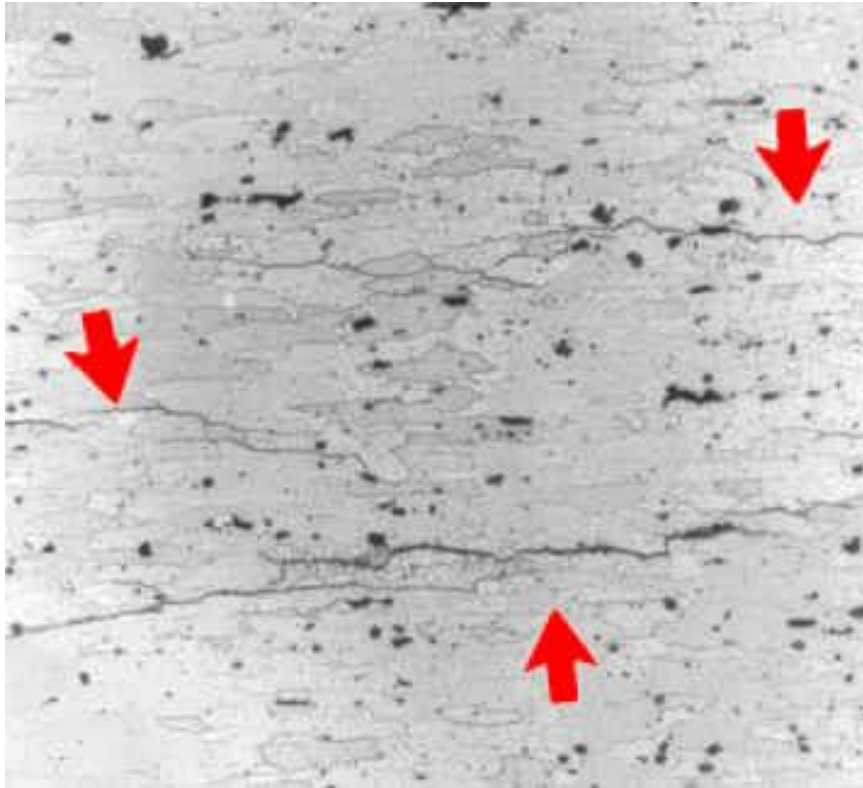
It is product of the attack located at the edges of the grain of the alloy.

It is a corrosion of origin electrolytic, due to the heterogeneity among the core (cathode) and the edges of the grain (anode).

Aluminium alloys in the cause of this corrosion is the presence of alloying precipitates along grain.



## Intergranular corrosion



## **Exfoliative corrosion**

It is a variety of intergranular corrosion occurring in rolled, extruded or forged products, where grains have been arranged in the direction in which the part has been worked in a layers disposition.

The corrosion at the grain edges produces the separation of the grain layers and so the delamination of the parts.

## Exfoliative corrosion



## Exfoliative corrosion



## Stress corrosion

In order to appear this type of corrosion it is necessary the simultaneous combination of four factors:

- Corrosive agent
- Susceptible material
- Stresses
- Time

It is a dangerous corrosive process by propagation (through a crack) and difficult to detect, because it is not visible, even after the discovery of the crack.

## Stress corrosion

Potential difference between different areas by heterogeneities that appear on the material caused by the stresses.

Stresses can come from directly applied loads or from the residual stresses that come from the manufacturing processes, especially by improper welds, forced assemblies, thermal treatments or drilling.

More likely manufacturing processes are: forged, extruded and sheets forming.

The parts subjected to residual stresses or preloads, as fittings lugs not protected against the bot torque of tightening are potential cases of stress corrosion.



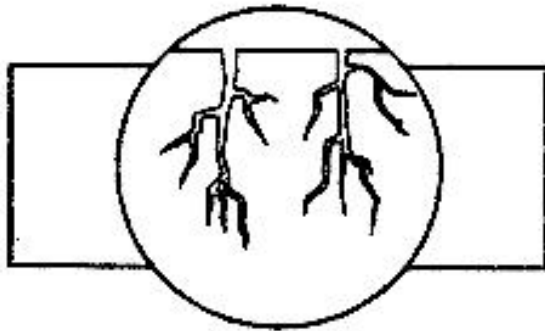
## **Stress corrosion**

Means to avoid the stress corrosion:

- Forming in annealed state
- Applying treatments against stress
- Machining before the final heat treatment
- Control the severity of temple minimizing internal tensions
- Shot peening to get surface compression stress
- Avoid residual stresses in assemblies, especially in bushes

## Stress corrosion

Corrosión por esfuerzo





## Fatigue corrosion

It is caused by the combined effect of cyclic loading and corrosion.

The failures occur by means of pitting or grooves that generates stress concentrations, and in a second phase the crack grows by effect of cyclic loading and corrosion.

Fatigue corrosion is prevented by:

- Eliminating the corrosive environment
- Properly machining and avoiding the grooves that produce dynamic stress concentration
- Designing with large fillet radii

## Fatigue corrosion



Aloha Airlines Flight 243 28/4/1988.  
Fatigue cracks found at skin lap joints



Corrosion at the lap joints lead to so called “pillowing”, whereby the faying surfaces are separated.

## Galvanic corrosion

When two metallic materials with an important difference in its electrochemical potential are in contact in the presence of an electrolyte always appears galvanic corrosion. The atmosphere moisture is enough to cause corrosion.

Which has more negative potential acts as anode and the most positive acts as cathode, the anode is corroded and the cathode remains unchanged.

The intensity of the attack depends on the potential difference of the elements and the areas rate. A small anode area and a large cathode area leads to a strong corrosion attack although the potential difference between the materials were small.

## Galvanic corrosion

The means to avoid the galvanic corrosion are:

- Use similar materials
- Modify surfaces potential
- Surfaces coating by noble metals (niquel, chromium)
- Interfay of non metallic materials: sealant, paints (on both surfaces or, al least, the cathodic surface).
- Interpose films ,parts or coatings less noble than the base metal, in order to be corroded instead of the part base material (sacrificial layer).
- Avoid electrolyte presence (sealing and draining)
- The fasteners must be of the same material as the parts to join o if it is not possible a more noble material and wet installed.

## Galvanic corrosion

Electrical bonding joints are very sensitive to galvanic corrosion. The damage is double because while some parts are corroded, the other parts are isolated.

Carbon fiber is a very noble material, therefore tends to create virulent galvanic pairs with most metals. Therefore protection films (fiberglass layers) are installed and the fixing parts used are A-286 stainless steel or titanium.



## Microbiologic corrosion (MIC)

It is defined as the deterioration of the materials caused by the presence and growth of microorganisms.

The MIC is generally produced in aqueous media where metals are submerged or floating. The biological organisms present in the water acting on the surface of the metal, speeding up the transport of oxygen to the metal surface, accelerating the corrosion process.

Microorganisms have a catalyst effect on the electrolytic corrosion:

- Bacteria and fungi produce areas of different aeration during their metabolic exchange with the environment, leading to the creation of cathodic areas
- They are able to destroy the surface protections and sealants
- They incapacitate some corrosion inhibitor products
- They produce changes in pH

**Microbiologic corrosion (MIC)**

Aluminum fuel tanks in wings and fuselage with incomplete drainage is attacked, causing leaks and malfunctions of the fuel system.

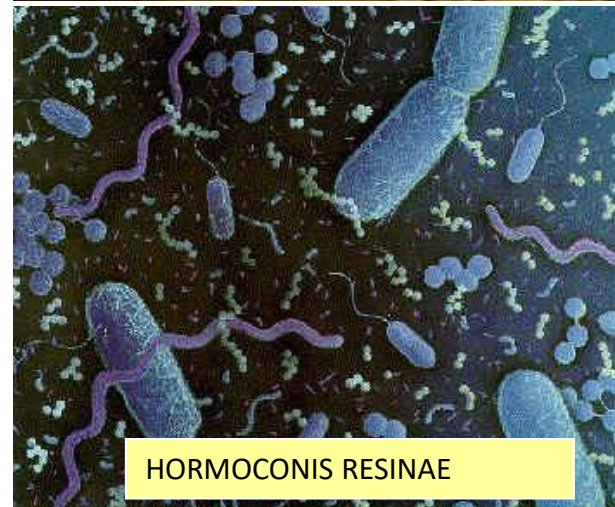
Bacteria enter with the air, water, dust, water filtered rainwater and additives that carries the fuel.

To protect the fuel tank, it is necessary to consider:

- Use the surface protectors with paints containing Cu y Zn
- Installing drainage systems in the lower areas
- Seal of joints



## Microbiologic corrosion (MIC)



HORMOCONIS RESINAE



## Filiform corrosion

It is a hybrid demonstration by intergranular corrosion and pitting.

Born in pitting of metal surfaces subjected to a moisture and Ionic atmosphere with presence of oxygen.

It takes place in the form of wires or mesh that grow in parallel to the surface. The meshes are propagated by an electrolytic process, in which the electrodes are created by the difference in oxygen concentration between the ends of the wire (CATHODE) and the adjacent surface (ANODE).

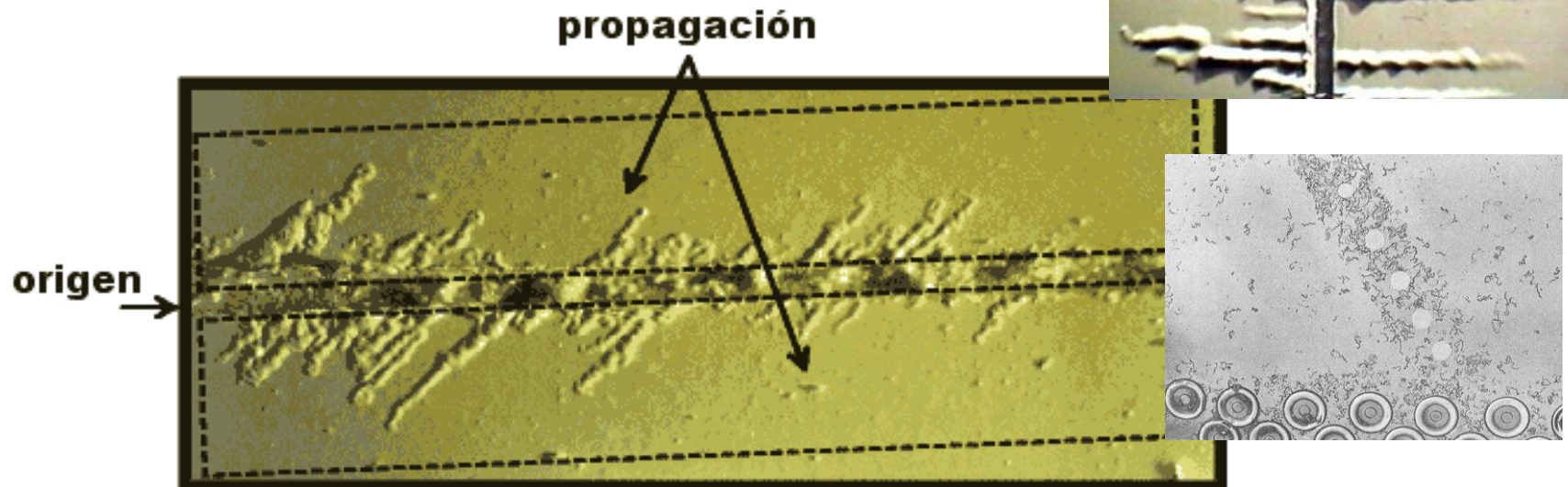
Generally appears in areas not protected:

- Holes
- The aluminum cladding elements attacked by pitting
- Surfaces with peeled paint

## Filiform corrosion

To avoid this kind of corrosion:

- Avoid moisture
- Use proper surface protection
- Use sealant at riveted joints



## **Decarburation**

The decarburation is the lost of Carbon content of steels, this lost is usually due to the corrosion of the steel, manufacturing processes which increase the part temperature in presence of oxygen boost the decarburation.

Typically the decarburation consist on perlite and cementite which decompose forming a layer of ferrite.

The consequences on the mechanical properties are reduction of hardness, mechanical strength, abrasion resistance and reduction of fatigue life.

## Aerospace materials

Aluminum alloys are the materials most used in military aircraft. In civilian aircraft more modern composite material already slightly overcomes metallic materials (A530 XWB: 53% composite material, 47% metallic material, 19% aluminum).

Besides the composite material and aluminum the most used materials in aeronautics are titanium alloys, steels, and alloys based on nickel and magnesium.

Copper and its alloys are used in electrical equipment.

Both cadmium and chromium are in process of elimination according to the REACH directive.

## Aluminum

Aluminum and its alloys can be affected by various types of corrosive attacks: pitting, exfoliation, intergranular, stress corrosion, etc.

The most effective protection is Anodizing, after the anticorrosive primer and the paint.

In most environments, the aluminum quickly forms an oxide film, waterproof and flexible, which protects the rest of the material in not excessively aggressive atmospheres.

If the layer of aluminum oxide is scratched, damaged or deteriorated should be restored, using Alodine and subsequently applying an organic protection (Primer).

## Aluminum

| MATERIAL |          |          | THERMAL TREATMENT | ROW MATERIAL     | CORROSION SENSIBILITY (2) |               |         |         |
|----------|----------|----------|-------------------|------------------|---------------------------|---------------|---------|---------|
| GERMANY  | USA      | FRANCE   |                   |                  | STRESS CORROSION          | INTERGRANULAR | PITTING | SURFACE |
| 3.1364   | 2024CLAD | 2024CLAD | T3/T4             | CLADDING         | NO                        | 3             | 3       | 3       |
|          |          |          | T351              |                  | NO                        | 3             | 2       | 1       |
| 3.1354   | 2024     | 2024     | T3                | SHEET            | NO                        | 1             | 1       | 2       |
|          |          |          | T351              |                  | SI (1)                    | 1             | 1       | 2       |
|          |          |          | T3/T4/T3510       | LAMINATED ROD    |                           |               |         |         |
|          |          |          | T3/T4/T3511       | EXTRUDED ROD     | NO                        | 1             | 1       | 2       |
|          |          |          | T3/T4/T6          | EXTRUDED TUBE    | SI (1)                    | 1             | 1       | 2       |
|          |          |          | T4/T3510/T3511    | EXTRUDED SECTION | NO                        | 2             | 3       | 3       |
| 3.3214   | 6061     | 6061     | T4/T6             | NON CLADDING ROD | NO                        | 2             | 3       | 3       |
|          |          |          |                   | EXTRUDED TUBE    | NO                        | 2             | 3       | 3       |
| 3.4144   | 7050     | 7050     | T76               | CLAD SHEET       | NO                        | 3             | 3       | 3       |
|          |          |          | T76               | SHEET            | NO                        | 2             | 2       | 2       |
|          |          |          | T7451             | PLATE            | NO (3)                    | 2             | 2       | 2       |
|          |          |          | T74511            | EXTRUDED ROD     |                           |               |         |         |
|          |          |          | T7452             | FORGING          |                           |               |         |         |
| 3.4377   | 7475     | 7475     | T761/T762         | CLAD SHEET       | NO                        | 3             | 3       | 3       |
|          |          |          |                   | SHEET            | NO                        | 2             | 2       | 2       |

(1) ST ONLY

(2) SENSIBILITY VALUES MEAN:

1: HIGH SENSIBILITY

2: MEDIUM SENSIBILITY

3: LOW SENSIBILITY

(3) FOR RESIDUAL STRENGTH BELOW 240 MPA

## Steel

The steel will corrode depending not only on the environment but the composition of the alloy and heat-treatment. There are elements such as copper, chromium, Silicon, phosphorus, which can alter corrosive behavior.

Passivation is a corrosion protection treatment that shall be applied to all stainless steel alloy.

Steels 17-4PH, 17-7PH, 15-5PH, AISI301, AISI304, AISI321 and AISI347 have acceptable corrosion behavior.

## Steel

| MATERIAL |          |                | THERMAL TREATMENT | ROW MATERIAL                                 | CORROSION SENSIBILITY (2) |               |         |         |
|----------|----------|----------------|-------------------|----------------------------------------------|---------------------------|---------------|---------|---------|
| GERMANY  | USA      | FRANCE         |                   |                                              | STRESS CORROSION          | INTERGRANULAR | PITTING | SURFACE |
| 1.6604   | -        | -              | T90               | RODS AND FORGING                             | NO                        | 3             | 2       | 1       |
|          | -        | 30NCD16        | T110              |                                              |                           |               |         |         |
|          | -        | -              | T125              |                                              |                           |               |         |         |
| 1.6747   | -        | 35NCD16        | T110              | RODS                                         | NO                        | 3             | 2       | 1       |
| 1.6404   | -        | 35NCD16        | T160              |                                              | YES                       | 3             | 2       | 1       |
| 1.5710   | -        | 35NC6          | T90               | RODS                                         | NO                        | 3             | 2       | 1       |
| 1.7734   | -        | 15CDV6         | T70               | SHEETS, RODS, FORGING AND TUBES              | NO                        | 3             | 2       | 1       |
|          |          |                | T100              |                                              |                           |               |         |         |
|          |          |                | T110              |                                              |                           |               |         |         |
| 1.4028   | -        | Z-30C13        | T90               | RODS                                         | NO                        | 3             | 2       | 3       |
| 1.4044   | AISI431  | Z-15CN17-03    | T90               | RODS AND FORGING                             | NO                        | 3             | 2       | 3       |
| 1.4306   | AISI304L | Z-3CN18        | SA                | SHEETS                                       | NO                        | 3             | 2       | 3       |
| 1.4314   | AISI304  | Z-6CN18-09     | H12               | SHEETS AND HYDRAULIC TUBES                   | NO                        | 3             | 2       | 3       |
| 1.4304   | AISI302  | Z-10CN18-09    | H38               | SPRINGS                                      | NO                        | 3             | 2       | 3       |
| 1.4544   | AISI321  | Z-10CNT18      | SA                | SHEETS, RODS, EXTRUSIONS AND HYDRAULIC TUBES | NO                        | 3             | 2       | 3       |
| 1.4546   | AISI347  | Z-6QNNb18-11   | SA                | RODS                                         | NO                        | 3             | 2       | 3       |
| 1.4125   | AISI4400 | Z-100CD17      | T130              | RODS                                         | NO                        | 3             | 2       | 3       |
| -        | AMS5561  | Z-4CMN21-09-06 | H16               | HYDRAULIC TUBES                              | NO                        | 3             | 2       | 3       |
| -        | -        | Z-8CND17-04    | T110              | RODS                                         | NO                        | 3             | 2       | 3       |

(1) ST ONLY

(2) SENSIBILITY VALUES MEAN:

- 1: HIGH SENSIBILITY
- 2: MEDIUM SENSIBILITY
- 3: LOW SENSIBILITY

(3) FOR RESIDUAL STRENGTH BELOW 240 MPA



## **Titanium**

Titanium and its alloys have good resistance to corrosion.

Titanium alloys have good galvanic compatibility with carbon fiber, can be used as fastener for this structures.

By its position in the electrochemical series, it is necessary to isolate it from aluminum, to prevent the corrosion of the aluminum.

## Titanium

| MATERIAL |         |        | THERMAL TREATMENT | ROW MATERIAL | CORROSION SENSIBILITY (2) |               |         |         |
|----------|---------|--------|-------------------|--------------|---------------------------|---------------|---------|---------|
| GERMANY  | USA     | FRANCE |                   |              | STRESS CORROSION          | INTERGRANULAR | PITTING | SURFACE |
| 3.7164   | AMS4911 | TA6V   | 0                 | SHEET        | NO                        | 3             | 3       | 3       |
|          | AMS4935 | TA6V   | 0                 | ROD          |                           |               |         |         |
|          | AMS4928 | TA6V   | 0                 | FORGING      |                           |               |         |         |
| 3.7184   | -       | -      | STA               | FORGING      |                           |               |         |         |
| 3.7134   | AMS4944 | -      | 0                 | TUBES        |                           |               |         |         |

(1) ST ONLY

(2) SENSIBILITY VALUES MEAN:

1: HIGH SENSIBILITY

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(3) FOR RESIDUAL STRENGTH BELOW 240 MPA

## **Corrosion protection. Index**

- Introduction
- Means to avoid corrosion
- Aircraft corrosion areas (environmental categories)
- Practical approach

## Corrosion protection. Introduction

Electrochemical corrosion process works as an electrical circuit, in which the intensity of the flux of electrons and ions is proportional to the difference in potential electric between poles and inversely proportional to the electrical resistance between the poles and between poles and electrolyte.

The protection means against corrosion act on variables involved in this circuit to reduce the intensity of corrosion.

## **Means to avoid corrosion.**

The means to avoid corrosion can be classified in four groups:

- I. Surface protections
- II. Painting
- III. Sealing
- IV. Draining

**Means to avoid corrosion. Surface protections**

Surface treatments that are used to prevent corrosion can be grouped into 4 types, taking into account the principle by which avoids corrosion:

- ANODIC PROTECTION
- CATHODIC PROTECTION
- ELECTROCHEMICAL POTENTIAL REDUCTION
- CLADDING

**Means to avoid corrosion. Surface protections**— ANODIC PROTECTION:

- It consists of forcing the generation on the anode of a surface layer of oxide which isolates the metal base, avoiding corrosion.
- This layer must be uniform, hermetic, with controlled thickness, hardness and adhesion to properly perform its function. This process is called passivation or anodized.
  - Chemical Process: Chemical Conversion Coating/alodine (Aluminum), passivate (Steel)
  - Electrochemical Process: Anodizing of Aluminum or Titanium (SAA, TSA, HSA, etc)

— CATHODIC PROTECTION:

Convert the anode to cathode protecting it with a layer of a less noble element (element with higher tendency to release  $e^-$ ), this protection is referred to as sacrificial layer.

**Means to avoid corrosion. Surface protections****— ELECTROCHEMICAL POTENTIAL REDUCTION:**

When the mating materials have a big potential difference a coating is applied to one of them, coating potential is similar to the less noble material so it is not corroded.

**— CLADDING:**

Is a pure aluminum coating, the absence of contaminants avoid the galvanic potential between the contaminants and the base material thus the corrosion is avoided until the part final protection is applied.



**Means to avoid corrosion. Painting:**

Within the term painting we speak mainly of 3 basic groups:

- Primer
- Paint
- External paint

**Primer:**

It is the first layer of paint applied to the parts after surface treatment.

There are many different types for different applications, its main functions are:

- Protection against corrosion.
- Protection against micro-organisms.
- Protection against lubricants, hydraulic fluids, fuel.
- Improve adhesion of paint.

**Means to avoid corrosion. Painting:****Paint, or top coat finish:**

Paint that is applied over the primer, depending on the base material and the area of the plane. There are many paint types with different functions:

- Protect primer from fluid or mechanical damage.
- Protection against lubricants, hydraulic fluids, fuel.
- Anti-static paint for discharge of static electricity.
- Resistance to abrasion.
- Resistance to UV radiation.
- Non-slip paint.
- Resistance to fire.
- Anti ice paint.
- Aerodynamic smoothing.
- Coating of fuel tanks: protects from corrosion, biological agents and serves as a barrier to fuel vapors

**Means to avoid corrosion. Painting:**

**External painting:**

On the outer faces of metal parts apply an anticorrosive primer and above it an exterior paint scheme, which can be applied to subassemblies or parts at final aircraft level.

The exterior paint scheme:

- Provide additional protection against corrosion in particular in the joining elements.
- Provide protection against the environment, including fluids, contaminants, temperature and UV radiation.
- Provide the A/C the customer distinctive.
- Should be chemically decapable.

Means to avoid corrosion. Painting:



**Means to avoid corrosion. Sealing:**

The correct application of the various types of sealing is one of the basic requirements to ensure adequate protection against corrosion, its main functions are:

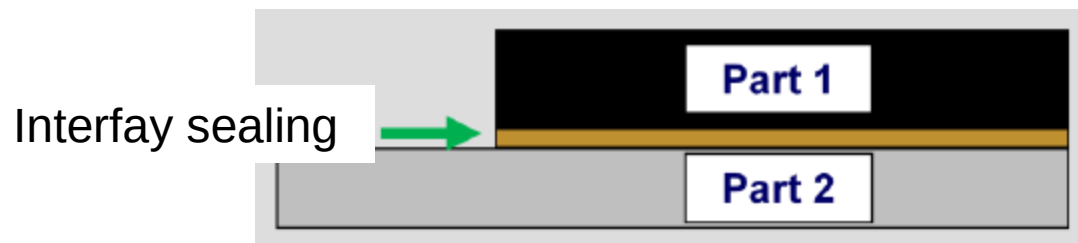
- Avoid ingestion of moisture.
- Prevent moisture buildup.
- Avoid contact between dissimilar materials.
- Avoid damage to the part surface treatment by friction.

Means to avoid corrosion. Sealing:

Interfay sealing:

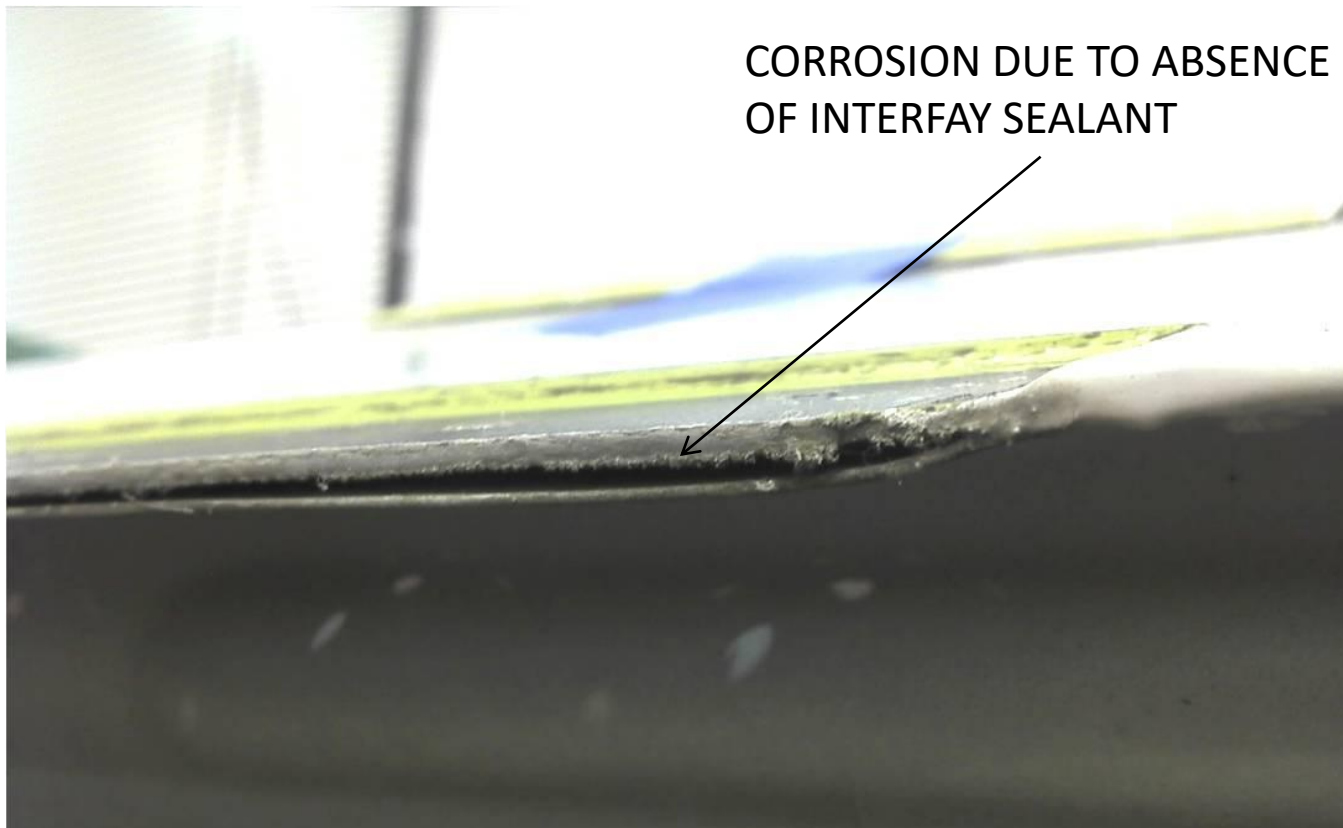
It consists of applying a coat of sealant between the faces of two parts. Main functions:

- Avoid income and accumulation of moisture between the parts
- Avoid direct contact between parts:
  - Reduce galvanic corrosion between them.
  - Avoiding the friction between parts due to deformations due to high stresses or vibration.
- Contribute to avoid the fuel tanks and pressure limits leakage



Means to avoid corrosion. Sealing:

Interfay sealing:



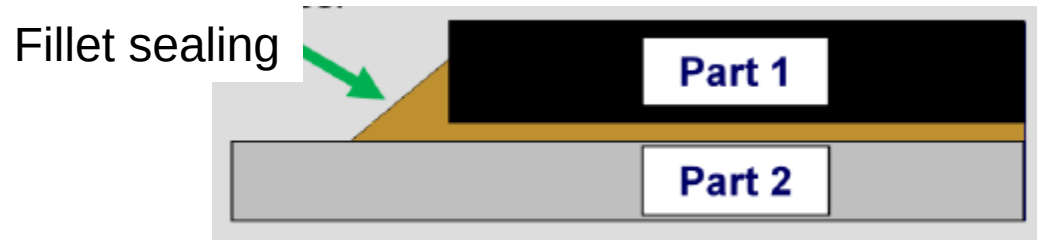
Means to avoid corrosion. Sealing:

Fillet sealing:

It consists of the application of a sealant bed on the contour of a part once installed on another by means of a sealant gun.

Main functions:

- Prevent entry of moisture between the parts.
- Contribute to avoid the fuel tanks and pressure limits leakage
- It prevents the accumulation of moisture on the edge of the part.
- Fill gaps and holes.





Means to avoid corrosion. Sealing:

Fillet sealing:



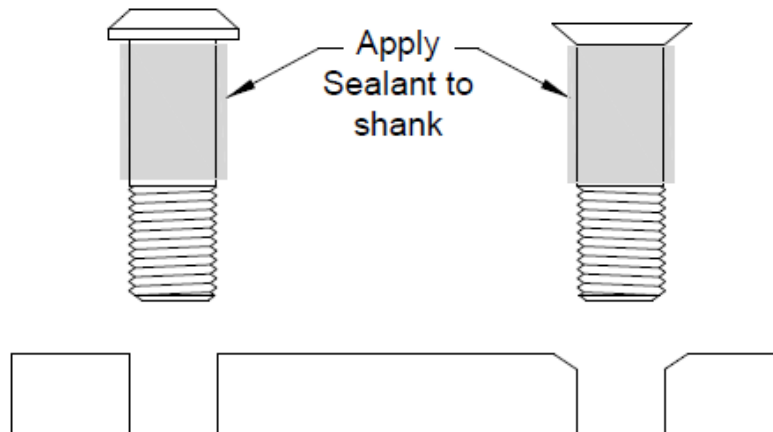
CORROSION DUE TO LACK OF SFILLET SEALANT  
IN COMBINATION WITH STRESSES.

Means to avoid corrosion. Sealing:

Wet install:

The fastener is impregnated by sealant previously to its installation, thus we get a sealant film between the fastener and the joined parts. Main functions:

- Avoid income and accumulation of moisture in the union.
- Avoid direct contact between the parts.
- Contribute to avoid the fuel tanks and pressure limits leakage



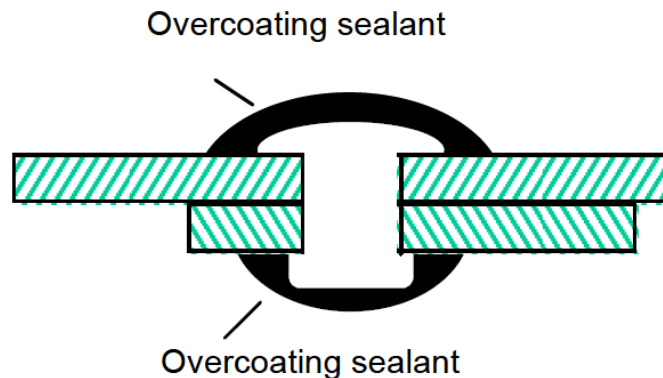
**Means to avoid corrosion. Sealing:**

## **Sealant overcoat:**

It consists of coating the head of an element of union (screw, rivet, etc) with sealant.

Main functions:

- Prevent entry of moisture in the union.
- Contribute to avoid the fuel tanks and pressure limits leakage

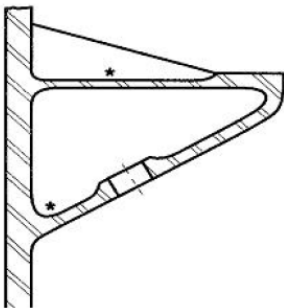


**Means to avoid corrosion. Draining:**

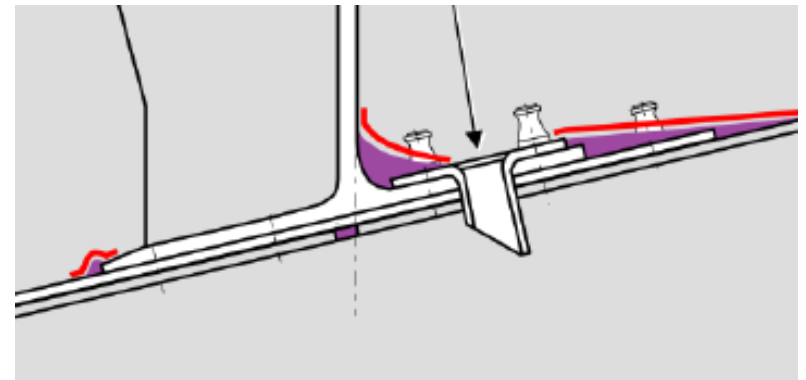
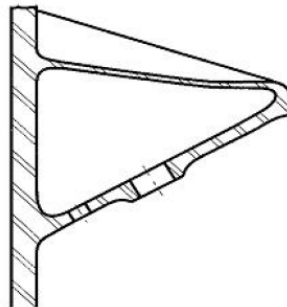
To prevent corrosion of the structures it is essential to avoid the accumulation of corrosive fluids in them, so it is necessary that the design provides a correct drainage of any possible fluid to the exterior of the aircraft.

Stringers, longerons, frames, etc are drained by holes 6 to 8.5 mm diameter, in order to guide the fluids to the lower part of the A/C symmetry plane where there are drain mast and drain valves adequately distributed.

Poor practice: water  
can be trapped at

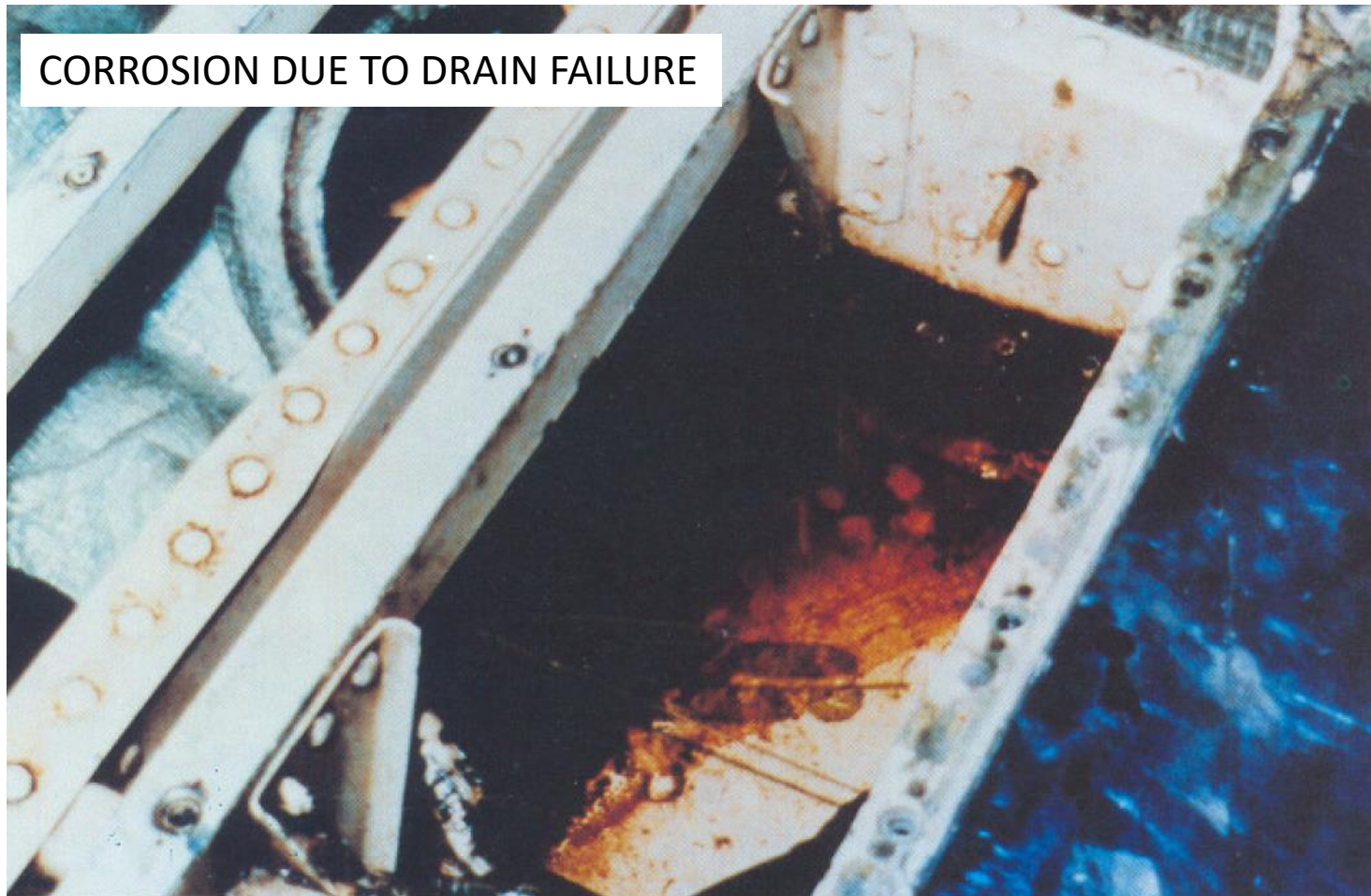


Good with slope  
and drain holes



Drain Mast

Means to avoid corrosion. Draining:

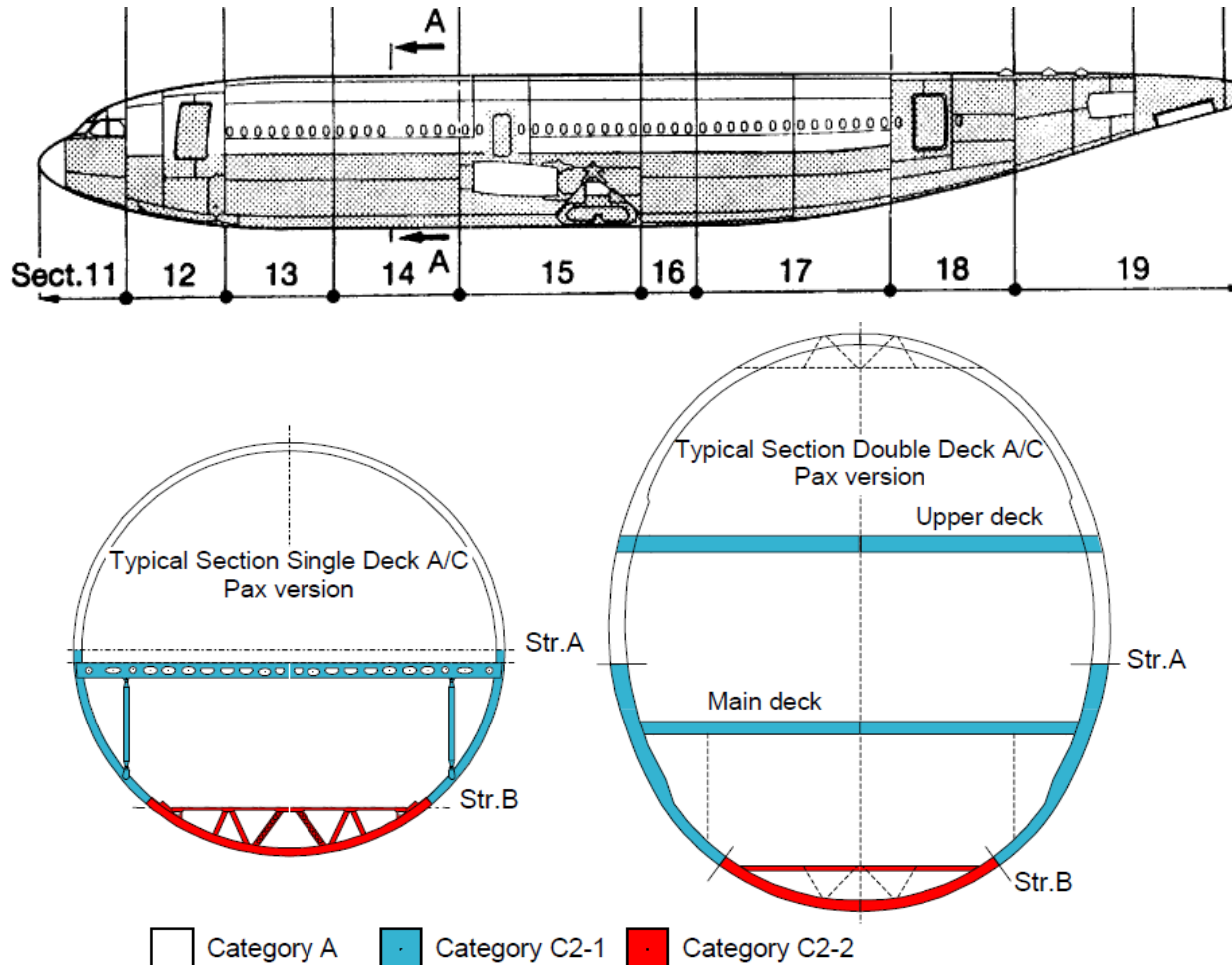


## Aircraft corrosion areas (environmental categories)

| Category |                                                                                                                                                                                                                                                                                                                       | Specific environment of the area                                                                                                  |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| A        |                                                                                                                                                                                                                                                                                                                       | Contact with air/condensation, easy accessible dry areas                                                                          |
| B        |                                                                                                                                                                                                                                                                                                                       | Contact with fuel (incl. pipes)                                                                                                   |
| C        | Areas: <ul style="list-style-type: none"><li>– which can be contaminated by: hydraulic fluid, toilet or galley liquids, water, lubricants, etc.</li><li>– in which condensation forms and/or is collected to a greater extent</li><li>– of difficult access</li><li>– with a high risk of accidental damage</li></ul> |                                                                                                                                   |
|          | C1                                                                                                                                                                                                                                                                                                                    | Areas in contact with water, moisture and occasional exposure to other liquids and/or areas with a high risk of accidental damage |
|          | C2-1                                                                                                                                                                                                                                                                                                                  | Areas in contact with water, moisture and frequent exposure to other liquids                                                      |
|          | C2-2                                                                                                                                                                                                                                                                                                                  | Areas in which water, moisture and other liquids can collect and areas of difficult access                                        |



## Aircraft corrosion areas (environmental categories)



Minimum requirements: Str.A - At least one stringer above floor level  
Str.B - Up to side & lower panel junction above cargo floor level

## **Practical approach to corrosion prevention. Index:**

- Surface protections selection
- Painting selection
- Sealing selection
- Fasteners protections
- Assemblies protections



**Practical approach to corrosion prevention:****Surface protections selection:**

In order to define the surface protection of a part we have to determine if there is any specific requirement for the part which leads to an specific protection, if not, the part generic protection is applied:

Generic protection. Depends on following factors:

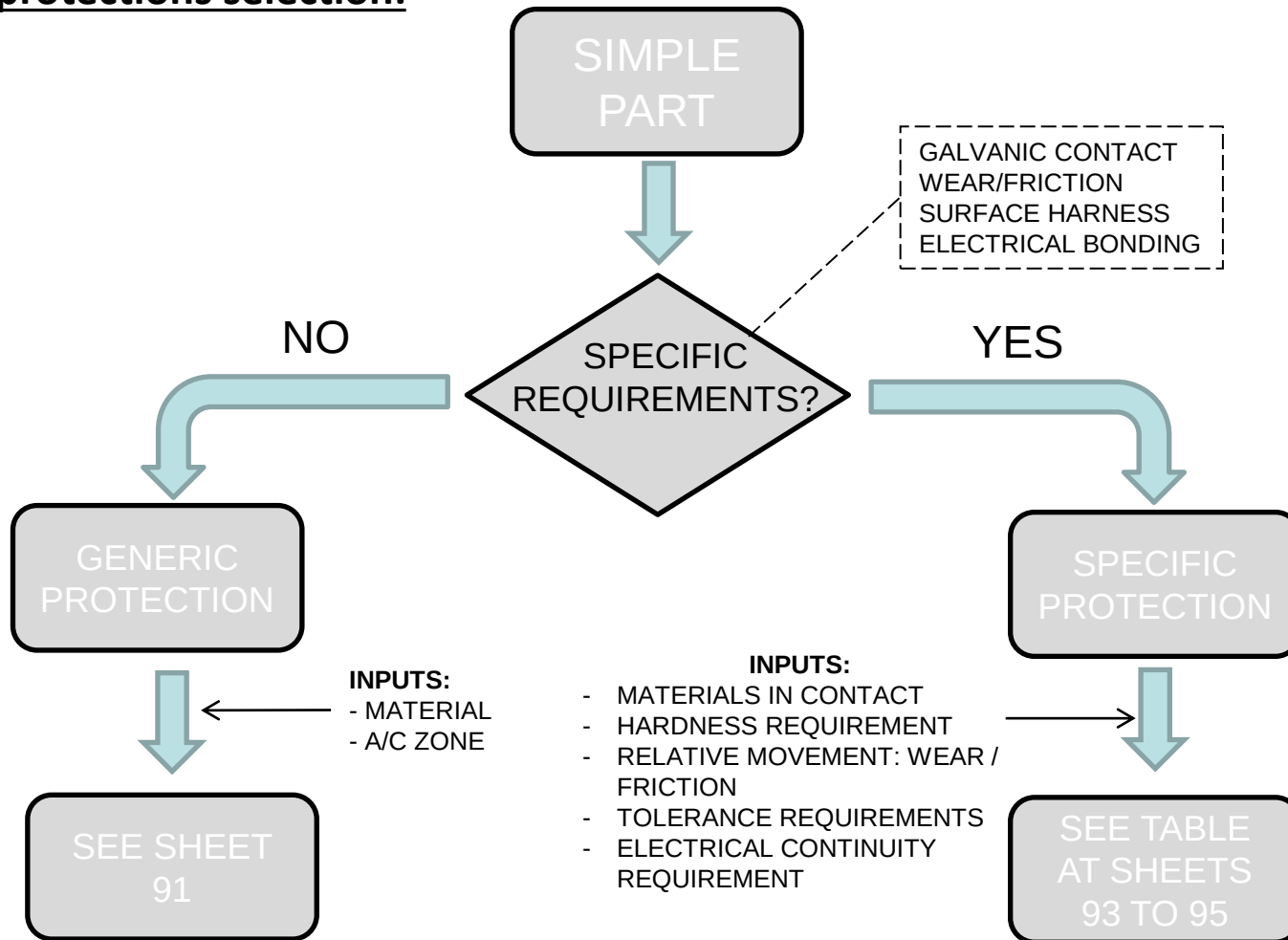
- Part raw material.
- A/C zone category where the part is installed.

Specific protection. Depends on the following factors:

- Mating parts material including the fasteners.
- Wear and friction protection
- Bonding
- Other functional requirements

## Practical approach to corrosion prevention:

### Surface protections selection:



## Practical approach to corrosion prevention:

**Surface protections:** Generic protection (see Refs. 7, 8 and 9)

| MATERIAL                                       | LOCATION                         | PROTECTIVE CODES                                                                                                                                                                                                        |
|------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aluminium Clad Sheets<br>Formed and Bent parts | Inside Fuel Tanks                | Chromic Acid Anodising Unsealed (55) + Primer (1)<br>Cod.450                                                                                                                                                            |
| Aluminium Clad Sheets<br>Formed and Bent parts | Outside fuel tank<br>Zone A      | Chemical Conversion Coating(60)+Primer(1)+Paint finish(3)<br>Cod.003                                                                                                                                                    |
|                                                | Outside fuel tank<br>Zone C      | Chromic Acid Anodising Unsealed (55) + Primer (1) + Paint Finish (3)<br>Cod.451                                                                                                                                         |
| Aluminium Clad Plates<br>Machined parts        | Inside Fuel Tanks                | Structures: Chromic Acid Anodising Unsealed (55) + Primer (1)<br>Pipes Fittings Internal Surface: Chemical Conversion Coating (60)<br>Pipes Fittings External Surface: Chemical Conversion Coating (60) +<br>Primer (1) |
| Aluminium Clad Plates<br>Machined parts        | Zone C and outside<br>fuel tank  | Chromic Acid Anodising Unsealed (55) + Primer (1) + Paint Finish (3)<br>Cod.451                                                                                                                                         |
| Alum. Extruded Profiles                        | Inside Fuel Tanks                | Chromic Acid Anodising Unsealed (55) + Primer (1)<br>Cod.450                                                                                                                                                            |
| Alum. Extruded Profiles                        | Outside Fuel Tanks               | Chromic Acid Anodising Unsealed (55) + Primer (1) + Paint Finish (3)<br>Cod.451                                                                                                                                         |
| Aluminium Fuel Pipes                           | Inside Fuel Tanks                | Internal Pipe Surface: Chemical Conversion Coating (60)<br>External Pipe Surface: Chemical Conversion Coating (60) + Primer (1)                                                                                         |
| Aluminium Pipes                                | Zone C and outside<br>fuel tank  | Chromate treated(60)+primer(1)+finish(3)<br>Cod.037                                                                                                                                                                     |
| Titanium hydraulic pipes                       | Any                              | Without protection<br>Cod.000                                                                                                                                                                                           |
| Stainless Steel                                | Outside Fuel Tanks               | Surface in contact with Aluminium: Cadmium Plating (50)<br>Free Surfaces: Passivation (68)<br>Surface in contact with Titanium: Blasting +Primer (1) + Finish (3)                                                       |
| Standard Rivet                                 | Zone C and Outside<br>fuel tanks | Flexible Varnish (11) or Primer (2) + Finish (3).<br>Cod.282                                                                                                                                                            |
| Electrical Bonding Areas in<br>Aluminium Parts | Any                              | Remove Existing Clad (Clad Sheets only) + Chemical Conversion<br>Coating (60)                                                                                                                                           |

## Practical approach to corrosion prevention:

### Surface protections: Specific protection

In order to know if it is necessary to apply any special protection to a part, we have to evaluate the part requirements and determine if any of the specific protections defined hereafter are adequate to comply with those requirements.

The part requirements can arise from the part functionality or from the part environment/assembly, the typical requirements that can lead to an specific protection are:

- Materials in contact
- Hardness requirement
- Relative movement: wear / friction
- Tolerance requirements
- Electrical continuity requirement

The following slides include tables with the description of the main generic and specific protections that can be applied, these tables define the protection characteristics and applicability parameters.

## Practical approach to corrosion prevention:

### Surface protections:

| MATERIAL | SURFACE TREATMENT                 | CLASSIFICATION                      | APPLICABILITY                                                                                                                                                                                                                                                                                             | COMMENTS                                                                                                            |
|----------|-----------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| ALUMINUM | CHROMIC ACID ANODIZING (CAA)      | ANODIC PROTECTION                   | REACH: FORBIDDEN DUE TO CHROME PRESENCE, SUSTITUDED BY TSA                                                                                                                                                                                                                                                | Th 5 $\mu\text{m}$<br>NON ELECTRICAL CONDUCTIVE<br>REDUCES FATIGUE RESISTANCE                                       |
|          | SULFURIC ACID ANODIZING (SAA)     | ANODIC PROTECTION / WEAR PROTECTION | PARTS UNDER FRICTION NON FATIGUE CRITICAL<br>CORROSION PROTECTION                                                                                                                                                                                                                                         | AFFECTS TOLERANCES (Th 5-15 $\mu\text{m}$ )<br>NON ELECTRICAL CONDUCTIVE                                            |
|          | HARD SULFURIC ANODIZING (HSA)     | ANODIC PROTECTION / WEAR PROTECTION | PARTS UNDER FRICTION AND/OR ABRASION WITHOUT<br>FATIGUE REQUIREMENTS                                                                                                                                                                                                                                      | AFFECTS TOLERANCES (Th 30-100 $\mu\text{m}$ )<br>NON ELECTRICAL CONDUCTIVE<br>FATIGUE RESISTANCE -50%               |
|          | TARTARIZ SULFURIC ANODIZING (TSA) | ANODIC PROTECTION                   | REACH: SUBSTITUTE OF CAA<br>BASIC ALUMINUM PARTS PROTECTION + PRIMER (ZONES A YB) OR PRIMER + PAINT (ZONE C)                                                                                                                                                                                              | Th 2-7 $\mu\text{m}$<br>NON ELECTRICAL CONDUCTIVE<br>REDUCES FATIGUE RESISTANCE                                     |
|          | CHEMICAL CONVERSION COATING (CCC) | ANODIC PROTECTION                   | PROTECTION FOR CLAD PARTS IN LOW CORROSIVE ZONES (A, C1)<br>CAN BE SUSTITUDED BY TSA + PRIMER (ZONES A AND B) OR PRIMER + PAINT (ZONE C)<br>MAINLY USED AT REPAIRS DUE TO EASY APPLICATION<br>STRUCTURAL AND FLUID SYSTEMS TUBES<br>BONDING CONTACT POINTS PROTECTION BECAUSE IT IS ELECTRICAL CONDUCTIVE | CORROSION PROTECTION LEVEL BELOW TSA<br>DOES NOT AFFECT TOLERANCES (Th < 2 $\mu\text{m}$ )<br>ELECTRICAL CONDUCTIVE |

## Practical approach to corrosion prevention: Surface protections:

| MATERIAL       | SURFACE TREATMENT             | CLASSIFICATION                        | APPLICABILITY                                                                                                                                                                                                                                                                                                                                              | COMMENTS                                                                           |
|----------------|-------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| STEEL AND CRES | PASSIVATE                     | ANODIC PROTECTION                     | CRES<br>CORROSION PROTECTION                                                                                                                                                                                                                                                                                                                               | DOES NOT AFFECT TOLERANCES<br>LOW ELEC. CONDUCTIVITY                               |
|                | CADMIUM PLATING / CADMIUM IVD | CATHODIC PROTECTION                   | STEEL, CRES AND COPPER ALLOY. STEEL CORROSION PROTECTION R<1510 MPa<br>CORROSION PROTECTION OF ALUMINIUM IN CONTACT WITH STEEL, CRES OR BRONZE (NOT APPLIED TO ALUMINIUM BUT TO MATING PART)<br>PLUS PRIMER (ZONES A YB) OR PRIMER + PAINT (ZONE C)<br>DO NOT APPLY TO AREAS UNDER WEAR.<br><b>REACH: PROTECTION FORBIDDEN SEE ALTERNATIVE PROTECTIONS</b> | Th 5-15 µm<br>ELECTRICAL CONDUCTIVE                                                |
|                | HARD CHROMIUM                 | CATHODIC PROTECTION / WEAR PROTECTION | PARTS UNDER WEAR NOR FATIGUE CRITICAL AND Rm <1450 Nmm <sup>2</sup><br><b>REACH: PROTECTION FORBIDDEN SEE ALTERNATIVE PROTECTIONS</b>                                                                                                                                                                                                                      | Th 50-200 µm<br>FATIGUE RESISTANCE -60%<br>HIGH SURFACE HARDNESS                   |
|                | NICKEL PHOSPHORUS - PTFE      | WEAR PROTECTION                       | Electroless (Chemical) Nickel Phosphorus - PTFE duplex coating<br>Medium contact pressure allowed: 300 MPa < pressure ≤ 500 MPa<br>Substrate: Nickel alloys, CRES, Low alloy steel<br><b>REACH: SUBSTITUTE OF HARD CHROMIUM</b>                                                                                                                            | Th negligible<br>HIGH SURFACE HARDNESS                                             |
|                | NICKEL PLATING                | CATHODIC PROTECTION / WEAR PROTECTION | PROTECTION AGAINST WEAR AND CORROSION                                                                                                                                                                                                                                                                                                                      | AFFECTS TOLERANCES (Th 20-50 µm)<br>ELECTRICAL CONDUCTIVE<br>HIGH SURFACE HARDNESS |
|                | PHOSPHATE PLATING             | CATHODIC PROTECTION / WEAR PROTECTION | STEEL. CRES R>1510 MPa<br>WELDED ASSEMBLIES<br>PLUS PRIMER + PAINT AT ZONE C (Zn-PHOSPHATING)<br>PROTECT PARTS WITH RELATIVE MOVEMENT, ADDING LUBRICANT (Mn-PHOSPHATING)                                                                                                                                                                                   | Th 5-10 µm<br>NON ELECTRICAL CONDUCTIVE                                            |
|                | SILVER PLATING                | FRICTION CORROSION PROTECTION         | STEEL, CRES AND COPPER ALLOY<br>APPLICABLE TO PART UNDER HIGH TEMPERATURE<br>HIGHLY RESISTANT TO WEAR AT HIGH TEMPERATURES                                                                                                                                                                                                                                 | Th 10 µm<br>ELECTRICAL CONDUCTIVE                                                  |
|                | ZINC SPRAY                    | CATHODIC PROTECTION                   | APPLICABLE TO STEEL, CRES, TITANIUM ALLOY AND BRONZE ALLOY<br><b>REACH: SUBSTITUTE OF CADMIUM PLATING/IVD ON CRES AND TITANIUM ALLOY</b><br>CORROSION PROTECTION                                                                                                                                                                                           | Th 4-15 µm                                                                         |
|                | ZINC-NICKEL PLATING           | CATHODIC PROTECTION                   | STEEL, CRES AND COPPER ALLOY<br><b>REACH: SUBSTITUTE OF CADMIUM PLATING OR CADMIUM IVD</b><br>DO NOT USE IN PARTS WITH Rm > 1450 Nmm <sup>2</sup><br>DO NOT APPLY TO AREAS UNDER WEAR                                                                                                                                                                      | Th 10-18 µm<br>ELECTRICAL CONDUCTIVE                                               |

## Practical approach to corrosion prevention: Surface protections:

| MATERIAL                    | SURFACE THREATMENT                                   | CLASIFICATION                                                                                                 | APPLICABILITY                                                                                                                                                                                                                                                                       | COMMENTS                                                       |
|-----------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| STEEL, CRES<br>AND TITANIUM | BRONZE SPRAY                                         | WEAR PROTECTION                                                                                               | WEAR PROTECTION                                                                                                                                                                                                                                                                     | Th 4 µm<br>ELECTRICAL<br>CONDUCTIVE                            |
|                             | ALUMINIUM SPRAY                                      | CATHODIC PROTECTION                                                                                           | REACH: SUSTITUTE OF CADMIUM PLATING OR<br>CADMIUM IVD<br>STEEL CATHODIC PROTECTION<br>TO PROTECT ALUMINUM IN CONTACT WITH STEEL<br>PRIMER NEEDED<br>DO NOT APPLY TO AREAS UNDER WEAR                                                                                                | Th 60-200 µm<br>ELECTRICAL<br>CONDUCTIVE                       |
|                             | ZINC SPRAY                                           | STEEL: CATHODIC PROTECTION<br>ALUMINIUM: ELECTROCHEMICAL POTENTIAL<br>REDUCTION WITH STEEL, CRES AND TITANIUM | REACH: SUSTITUTE OF CADMIUM PLATING/IVD ON<br>CRES AND TITANIUM ALLOY<br>STEEL CATHODIC PROTECTION<br>TO PROTECT ALUMINUM IN CONTACT WITH STEEL,<br>CRES OR TITANIUM                                                                                                                | Th 40 µm<br>ELECTRICAL<br>CONDUCTIVE                           |
|                             | ALUMINIUM IVD                                        | ELECTROCHEMICAL POTENTIAL REDUCTION<br>BETWEEN TITANIUM AND ALUMINIUM                                         | REACH: SUSTITUTE OF CADMIUM PLATING OR<br>CADMIUM IVD<br>ALUMINUM GALVANIC PROTECTION<br>DO NOT APPLY TO AREAS UNDER WEAR                                                                                                                                                           | Th 25-50 µm (4-12 µm<br>FASTENERS)<br>ELECTRICAL<br>CONDUCTIVE |
|                             | TITANIUM NITRIDE (TiN)                               | WEAR PROTECTION                                                                                               | REACH: SUSTITUTE OF HARD CHROMIUM<br>APPLICABLE TO CRES, NICKEL AND TITANIUM<br>WEAR PROTECTION                                                                                                                                                                                     | Th 2-5 µm<br>VERY HIGH SURFACE<br>HARDNESS                     |
|                             | CARBURO DE TUNGTEÑO (WC<br>THERMAL SPRAY, I+D-P-375) | WEAR PROTECTION                                                                                               | REACH: SUSTITUTE OF HARD CHROMIUM<br>APPLICABLE TO CRES, NICKEL AND TITANIUM<br>WEAR PROTECTION IN JOINTS:<br>- TITANIUM BOLT/BRONZE BUSHING<br>- CRES BOLT/BRONZE BUSHING<br>- threads (thickness 1.5 to 3 µm)<br>Medium contact pressure allowed: 300 MPa <<br>pressure ≤ 500 Mpa | Th 5 µm<br>ELECTRICAL<br>CONDUCTIVE                            |
| TITANIUM                    | ANODIZADO SULFURICO (SAA)                            | PROVIDES ADHERENCE FOR PRIMER OR SOLID FILM<br>LUBRICANT                                                      | ADHESION FOR SOLID FILM LUBRICANT<br>DO NOT APPLY TO EXTERNAL SURFACES                                                                                                                                                                                                              | Th < 1 µm<br>LOW ELECTRICAL<br>CONDUCTIVITY                    |

## Practical approach to corrosion prevention:

**Painting selection:** Usually defined at the same table defining the surface protections  
(See table at next slide)

Aluminum parts: Aluminum parts are painted after receiving the surface protection, the basic scheme is as follows:

Zone A: Epoxy primer

Zone B: Antibacterial primer

Zone C: Epoxy primer + polyurethane top coat

CRES parts: Painting depends on the A/C program

Titanium parts: Do not require painting



## Practical approach to corrosion prevention:

### Painting selection:

| MATERIAL                                       | LOCATION                         | PROTECTIVE CODES                                                                                                                                                                                                        |
|------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aluminium Clad Sheets<br>Formed and Bent parts | Inside Fuel Tanks                | Chromic Acid Anodising Unsealed (55) + Primer (1)<br>Cod.450                                                                                                                                                            |
| Aluminium Clad Sheets<br>Formed and Bent parts | Outside fuel tank<br>Zone A      | Chemical Conversion Coating(60)+Primer(1)+Paint finish(3)<br>Cod.003                                                                                                                                                    |
|                                                | Outside fuel tank<br>Zone C      | Chromic Acid Anodising Unsealed (55) + Primer (1) + Paint Finish (3)<br>Cod.451                                                                                                                                         |
| Aluminium Clad Plates<br>Machined parts        | Inside Fuel Tanks                | Structures: Chromic Acid Anodising Unsealed (55) + Primer (1)<br>Pipes Fittings Internal Surface: Chemical Conversion Coating (60)<br>Pipes Fittings External Surface: Chemical Conversion Coating (60) +<br>Primer (1) |
| Aluminium Clad Plates<br>Machined parts        | Zone C and outside<br>fuel tank  | Chromic Acid Anodising Unsealed (55) + Primer (1) + Paint Finish (3)<br>Cod.451                                                                                                                                         |
| Alum. Extruded Profiles                        | Inside Fuel Tanks                | Chromic Acid Anodising Unsealed (55) + Primer (1)<br>Cod.450                                                                                                                                                            |
| Alum. Extruded Profiles                        | Outside Fuel Tanks               | Chromic Acid Anodising Unsealed (55) + Primer (1) + Paint Finish (3)<br>Cod.451                                                                                                                                         |
| Aluminium Fuel Pipes                           | Inside Fuel Tanks                | Internal Pipe Surface: Chemical Conversion Coating (60)<br>External Pipe Surface: Chemical Conversion Coating (60) + Primer (1)                                                                                         |
| Aluminium Pipes                                | Zone C and outside<br>fuel tank  | Chromate treated(60)+primer(1)+finish(3)<br>Cod.037                                                                                                                                                                     |
| Titanium hydraulic pipes                       | Any                              | Without protection<br>Cod.000                                                                                                                                                                                           |
| Stainless Steel                                | Outside Fuel Tanks               | Surface in contact with Aluminium: Cadmium Plating (50)<br>Free Surfaces: Passivation (68)<br>Surface in contact with Titanium: Blasting +Primer (1) + Finish (3)                                                       |
| Standard Rivet                                 | Zone C and Outside<br>fuel tanks | Flexible Varnish (11) or Primer (2) + Finish (3).<br>Cod.282                                                                                                                                                            |
| Electrical Bonding Areas in<br>Aluminium Parts | Any                              | Remove Existing Clad (Clad Sheets only) + Chemical Conversion<br>Coating (60)                                                                                                                                           |

## **Practical approach to corrosion prevention:**

### **Painting selection:**

#### CFRP parts:

##### Internal surfaces:

In general no need to paint since the carbon is a corrosion resistant material.

At the inner side of the sandwich parts a layer of Tedlar is applied in order to avoid moisture inlet.

Only parts with specific requirements to be frequently disassembled, in this case primer (Zone A) or primer + finish (Zone C) is applied.

##### External surfaces:

No composite part is painted

## Practical approach to corrosion prevention:

### Painting selection:

Primer or paint shall not be applied in the following cases:

- Mating surfaces, moving or mechanically adjusted parts.
- Threads or screws.
- Parts covered by grease, lubricants or hydraulic oil.
- Close tolerance fits, holes for inserts and lubrication holes.
- Electrical connections
- Elastomer parts.
- Bearing holes.
- Oxygen pipes inner side.
- Part which will suffer temperatures above:

| PAINT                           | MAX TEMP<br>(°c) |
|---------------------------------|------------------|
| Epoxy primer                    | 150              |
| Polyurethane varnish            | 150              |
| Antierosion paints              | 150              |
| Silicon paints aluminum charged | 600              |

## **Practical approach to corrosion prevention:**

### **Sealing selection:**

Interfay: Applied in following cases:

- Corrosion critical areas (Zone C2), except clip-frame and clip-stringer joint
- Part joined to pressure boundaries
- Dissimilar materials
- Fatigue loaded joints
- Part joined to fuel tank boundaries
- Bushings and bearings installation

## Practical approach to corrosion prevention:

### Sealing selection:

Fillet seal: Applied in following cases:

- Dissimilar materials
- Fuel tank boundary (inside zone B only)
- Part joined to pressure boundaries
- Bushings and bearings installation
- Corrosion critical areas (zone C2), except clips-frame and clip-stringer joint

## Practical approach to corrosion prevention:

### Sealing selection:

Wet install: Applied in following cases:

- Dissimilar materials
- Part joined to fuel tank boundaries
- Part joined to pressure boundaries
- Bushings and bearings installation
- Corrosion critical areas:
  - CRES: Zone C2
  - Aluminum: Zones C1 and C2

## **Practical approach to corrosion prevention:**

### **Sealing selection:**

Overcoat: Applied in following cases:

- Dissimilar materials
- Part joined to fuel tank
- Bonding point at corrosion critical areas (Zone C)
- Corrosion critical areas (Zone C2: lavatories, galleys, bilge)

## Practical approach to corrosion prevention:

### Sealing selection:

|                                                         | INTERFAY SEALANT | FILLET SEALANT   | WET INSTALL | SEALANT OVERCOATING        |
|---------------------------------------------------------|------------------|------------------|-------------|----------------------------|
| Dissimilar Materials                                    | X                | X                | X           | X (Ref.4 Paragraph 9.1.1)  |
| Joints to pressure boundaries                           | X                | X                | X           |                            |
| Joints to fuel tank boundaries                          | X                | X                | X           | X                          |
| Bushings and Bearings installation                      | X                | X                | X           |                            |
| Corrosion Critical areas C2-1 and C2-2                  | X (Except clips) | X (Except clips) | X           | Lavatories, Galleys, bilge |
| Bonding point at Corrosion Critical areas C2-1 and C2-2 |                  |                  |             | X                          |

This sealing chart is for reference only, for each program the corresponding corrosion protection requirements should be reviewed.



## Practical approach:

### Fasteners protection : Bolt selection

|    |                                                                         |
|----|-------------------------------------------------------------------------|
|    | galvanic incompatible                                                   |
| 1) | Zn-plating shall not be used in contact with fuel                       |
| 2) | Combination to be avoided, because IVD-Al may be dissolved by corrosion |
| 3) | washer recommended                                                      |
| 4) | Combination to be avoided, because Cd will corrode                      |
| 5) | Combination to be checked case by case, because Cd will corrode         |

| Bolts (parts, which may go through the hole of a part like bolts, rivets, bearings, bushes, inserts) |         |        | Structure substrate |              |       |    |      |                    |    |    |              |
|------------------------------------------------------------------------------------------------------|---------|--------|---------------------|--------------|-------|----|------|--------------------|----|----|--------------|
|                                                                                                      |         |        | Al                  | Ti           | Cres  | St | CFRP | GFRP AFRP Plastics | Cu | Ni |              |
| Substrate                                                                                            | Ti      | Finish | none                |              | ✓     | ✓  |      | ✓                  | ✓  | ✓  | ✓            |
|                                                                                                      |         |        | RBC                 | ✓            | ✓     | ✓  | ✓    | ✓                  | ✓  | ✓  | ✓            |
|                                                                                                      |         |        | SAA                 | ✓            | ✓     | ✓  | ✓    | ✓                  | ✓  | ✓  | ✓            |
|                                                                                                      |         |        | WC-C:H              |              | ✓     | ✓  |      | not assessed       | ✓  | ✓  | not assessed |
|                                                                                                      |         |        | CuNiIn              |              | ✓     | ✓  |      | not assessed       | ✓  | ✓  | ✓            |
|                                                                                                      |         |        | IVD-Al              | ✓            | 2)    | 2) | 2)   |                    | ✓  | 2) | 2)           |
|                                                                                                      |         |        | Ph-Fluorid          | ✓            | ✓     | ✓  |      | ✓                  | ✓  | ✓  | ✓            |
|                                                                                                      | Cres    |        | none                |              | ✓     | ✓  |      | ✓                  | ✓  | ✓  | ✓            |
|                                                                                                      |         |        | Passivated          |              | ✓     | ✓  |      | ✓                  | ✓  | ✓  | ✓            |
|                                                                                                      |         |        | RBC                 | ✓            | ✓     | ✓  | ✓    | ✓                  | ✓  | ✓  | ✓            |
|                                                                                                      |         |        | IVD-Al              | ✓            | 2)    | 2) | 2)   |                    | ✓  | 2) | 2)           |
|                                                                                                      |         |        | Zn-plated 1)        | ✓            |       |    | ✓    |                    | ✓  | ✓  |              |
|                                                                                                      |         |        | ZnNi-plated         | ✓            |       |    | ✓    |                    | ✓  |    |              |
|                                                                                                      |         |        | WC-C:H              |              | ✓     | ✓  |      | not assessed       | ✓  | ✓  | not assessed |
|                                                                                                      | St      |        | Hard-Cr             | not assessed | ✓     | ✓  |      | ✓                  | ✓  | ✓  | ✓            |
|                                                                                                      |         |        | Cd-plated           | ✓            | 3) 5) |    | 4)   | 5)                 | ✓  |    | 5)           |
|                                                                                                      |         |        | Cd-plated           | ✓            | 3)    | 4) | ✓    |                    | ✓  |    |              |
|                                                                                                      |         |        | Hard-Cr             | not assessed | ✓     | ✓  |      | ✓                  | ✓  | ✓  | ✓            |
|                                                                                                      |         |        | ZnNi-plating        | ✓            |       | ✓  | ✓    |                    | ✓  |    |              |
|                                                                                                      | Al      |        | Zn-plated 1)        | ✓            |       |    | ✓    |                    | ✓  | ✓  |              |
|                                                                                                      |         |        | CCC                 | ✓            |       |    |      |                    | ✓  |    |              |
|                                                                                                      |         |        | Anod                | ✓            |       |    |      |                    | ✓  |    |              |
|                                                                                                      | Ni-base |        | none                |              | ✓     | ✓  |      | ✓                  | ✓  | ✓  | ✓            |
|                                                                                                      |         |        | RBC                 | ✓            | ✓     | ✓  | ✓    | ✓                  | ✓  | ✓  | ✓            |
|                                                                                                      |         |        | IVD-Al              | ✓            | 2)    | 2) | 2)   |                    | ✓  | 2) | 2)           |
|                                                                                                      |         |        | Passivated          |              | ✓     | ✓  |      | ✓                  | ✓  | ✓  | ✓            |
|                                                                                                      |         |        | Cd-plated           | ✓            | 3)    |    | ✓    |                    | ✓  |    |              |

## Practical approach to corrosion prevention:

## Fasteners protection : Nut selection

[illegible]

|    |                                                   |
|----|---------------------------------------------------|
| 1) | Zn-plating shall not be used in contact with fuel |
| 2) | after wet or dry blasting or sanding              |

## Practical approach to corrosion prevention:

### Fasteners protection : Bolt/Nut allowed couples

| Fastener<br>(bolts, rivets, bearings, bushes, nuts, collars, washer) |          |              |              |               |        | Substrate     |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|----------------------------------------------------------------------|----------|--------------|--------------|---------------|--------|---------------|-----|-------------|--------------|-------------|-----|--------|--------------|---------------|---------|----------|------|---------------|------|-----|--------|-----------|------|
|                                                                      |          |              |              |               |        | Ti            |     | Cres        |              |             | St  |        | Al           |               | Ni      |          |      |               | Cu   |     | Co     | Plastics  |      |
|                                                                      |          |              |              |               |        | Finish        |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          |              |              |               |        | none          | RBC | SAA         | IVD-Al       | none        | RBC | IVD-Al | Zn-plated 1) | Cd-plated     | Hard-Cr | PA Nylon | none | Anod          | none | RBC | IVD-Al | Cd-plated | none |
| Substrate                                                            | Ti       | WC-C:H       |              |               |        | Pass          |     | ZnNi-plated | Zn-plated 1) |             |     | CCC    |              | Pass          |         |          |      | Ni-plated     |      |     |        |           |      |
|                                                                      |          | CuNiIn       |              |               |        | Silver-plated |     |             | Cd-plated    | ZnNi-plated |     |        |              | Silver-plated |         |          |      | Silver-plated |      |     |        |           |      |
|                                                                      |          | PhosFl       |              |               |        | WC-C:H        |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          |              |              |               |        | Hard-Chrome   |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      | Cres     | none         | WC-C:H       | CuNiIn        | PhosFl |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | RBC          |              |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | SAA          |              |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | IVD-Al       |              |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      | St       | None         | Passivated   | Silver-plated | WC-C:H | Hard-Chrome   |     |             |              | 2)          |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | RBC          |              |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | IVD-Al       |              |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | Zn-plated 1) | ZnNi-plated  | Cd-plated     |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      | Al       | Cd-plated    | Zn-plated 1) | ZnNi-plated   |        |               | 2)  |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | Hard-Chrome  |              |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | PA, Nylon    |              |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | none         | CCC          |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      | Ni       | Anod         |              |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | none         | Passivated   | Silver-plated |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | RBC          |              |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | IVD-Al       |              |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      | Cu       | Cd-plated    |              |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | none         | Ni-plated    | Silver-plated |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | ZnNi-plated  |              |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      |          | none         |              |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |
|                                                                      | Plastics | Zn-plated 1) |              |               |        |               |     |             |              |             |     |        |              |               |         |          |      |               |      |     |        |           |      |

✓

galvanical compatible

galvanically incompatible

1) Zn-plating shall not be used in contact with fuel

2) Cd-plated St/Cres, pass: Combination to be avoided, because Cd will corrode

## Practical approach to corrosion prevention:

### Fasteners protection : Fasteners line protection at assembly (Generic)

| ENVIRONMENTAL CATEGORY                                                     | FASTENER MATERIAL  | AL OR ST                                   |                    |    |              | TI           |                    |    |              | CRES OR NI   |            |    |                        |              |
|----------------------------------------------------------------------------|--------------------|--------------------------------------------|--------------------|----|--------------|--------------|--------------------|----|--------------|--------------|------------|----|------------------------|--------------|
|                                                                            | STRUCTURE MATERIAL | AL OR AL-FML                               | TI OR CRES OR CFRP | ST | GFRP OR AFRP | AL OR AL-FML | TI OR CRES OR CFRP | ST | GFRP OR AFRP | AL OR AL-FML | TI OR CRES | ST | CFRP                   | GFRP OR AFRP |
| A                                                                          |                    | 2                                          |                    | 2  | 2            | 2            | 1                  | 2  | 1            | 2            | 1          | 2  | 1                      | 1            |
| B                                                                          |                    | 4                                          |                    | 4  | 4            | 4            | 1                  | 4  | 1            | 4            | 1          | 4  | 1                      | 1            |
| B (tank boundaries)                                                        |                    | 6                                          |                    |    |              | 6            | 6                  |    |              | 6            | 6          |    | 6                      |              |
| C2-1                                                                       |                    | 3                                          |                    | 3  | 3            | 3            | 1                  | 3  | 1            | 3            | 1          | 3  | 1                      | 1            |
| C2-2<br>Also external areas hidden by fairings                             |                    | 8                                          |                    | 8  | 8            | 8            | 1                  | 8  | 1            | 8            | 1          | 8  | 1 for Cres<br>8 for Ni | 1            |
| External Areas, not hidden by fairings (no classification into categories) |                    | 1 before application of final paint system |                    |    |              |              |                    |    |              |              |            |    |                        |              |

#### Re-preservation schemes

|   |                                                                                                                       |
|---|-----------------------------------------------------------------------------------------------------------------------|
| 1 | No preservation                                                                                                       |
| 2 | P<br>or Cidap<br>or Varnish                                                                                           |
| 3 | P + Tc<br>or P + Varnish<br>or Cidap                                                                                  |
| 4 | P<br>or P + Tc<br>or Sealant                                                                                          |
| 5 | P + Tc<br>or P + Cidap<br>or P + Varnish<br>or Sealant + P + Tc<br>or Sealant + P + Cidap<br>or Sealant + P + Varnish |
| 6 | Sealant overcoating                                                                                                   |

|    |                                                                                            |
|----|--------------------------------------------------------------------------------------------|
| 7  | P + Tc<br>or P + Varnish<br>or Varnish<br>or Cidap                                         |
| 8  | P + Tc<br>or P + Varnish<br>or P + Cidap                                                   |
| 9  | P<br>or P + Tc                                                                             |
| 10 | Sealant + P + Tc<br>or Sealant + Varnish<br>or Sealant + Cidap                             |
| 11 | Sealant + P + Tc<br>or Sealant + P + Varnish<br>or Sealant + Varnish<br>or Sealant + Cidap |
| 12 | Sealant + P<br>or Sealant + P + Tc                                                         |

| Protection level                                                                  | Protection system                            |
|-----------------------------------------------------------------------------------|----------------------------------------------|
| <div style="text-align: center;">             high<br/>↑<br/>low           </div> | Sealant + Primer + Topcoat/Sealant + Varnish |
|                                                                                   | Sealant + Primer                             |
|                                                                                   | Sealant                                      |
|                                                                                   | Primer + Topcoat/Primer + Cidap              |
|                                                                                   | Primer                                       |
|                                                                                   | Cidap                                        |
|                                                                                   | No preservation                              |

## Practical approach to corrosion prevention:

### Fasteners protection : Fasteners line protection at assembly (Specific for Al / CFRP)

Table 43: Fastener Line Protection on Al/conductive plastics (e.g. CFRP) joints

| Environmental Category | Preformed head      |                       | Collar              |              |                       |    |
|------------------------|---------------------|-----------------------|---------------------|--------------|-----------------------|----|
|                        | Ti or Cres bolt     |                       | Al                  | Cres* or Ti* | Cres                  | Ti |
|                        | On Al side of joint | On CFRP side of joint | On Al side of joint |              | On CFRP side of joint |    |
| A                      | c                   | a                     | c                   | c            | a                     | a  |
| B (internal)           | h                   | a                     | h                   | h            | h                     | a  |
| B (boundary)           | k                   | f                     | k                   | k            | k                     | f  |
| C1                     | g                   | a                     | c                   | c            | c                     | a  |
| C2-1                   |                     |                       |                     |              |                       |    |
| C2-2                   | e                   | c                     | i                   | i            | i                     | i  |

\* Depending on collar surface protection additional washer might be necessary

In case of Al/CFRP assemblies in cat. C2-2 areas, the re-preservation of the fasteners by varnish only is not permissible; primer or sealant always needs to be applied first.

#### Fastener line protection schemes

| Code letter | Protection scheme                                    | Code letter | Protection scheme       |
|-------------|------------------------------------------------------|-------------|-------------------------|
| a           | No preservation                                      | f           | S over coating          |
| b           | P<br>or Cidap                                        | g           | P + Tc<br>or P + Cidap  |
| c           | P + Tc<br>or P + Cidap<br>or Cidap                   | h           | P<br>or P + Tc          |
| d           | P<br>or P + Tc<br>or S over coating                  | i           | S + P + Tc<br>or S + V* |
| e           | P + Tc<br>or P + Cidap<br>or S + P + Tc<br>or S + V* | k           | S + P<br>or S + P + Tc  |

#### Legend

|       |                                                                           |
|-------|---------------------------------------------------------------------------|
| P     | Primer                                                                    |
| Tc    | Topcoat                                                                   |
| Cidap | Corrosion inhibiting direct adhesion paint (for fastener line protection) |
| S     | Sealant                                                                   |
| V     | Varnish (Sealant bead protection coating)                                 |

\* Varnish to be used in hydraulic fluid area only

## Practical approach:

### Summary : Similar materials

| SIMILAR MATERIAL ASSEMBLY AND FREE SURFACES REQUIREMENTS |                       |                            |  |                                                                                                                                                     |                                                                                  |                        |
|----------------------------------------------------------|-----------------------|----------------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------|
| CAT.<br>(Chap.<br>7.1)                                   | MATERIAL<br>OF PART   | SURFACE<br>PROTECTION      |  | INTERFACE AND ASSEMBLY                                                                                                                              | POST-ASSEMBLY<br>FASTENER HEAD<br>AND NUT                                        | HYDRA<br>ULIC<br>AREAS |
| A                                                        | Aluminium<br>(clad)   | TSA <sup>1</sup> + P       |  | Wet installation of fastener <sup>2</sup>                                                                                                           | P<br>or V<br>or CI DAP                                                           |                        |
|                                                          | Aluminium<br>(unclad) | TSA + P                    |  | Interfay sealant if attachment to skin on pressurized areas only                                                                                    |                                                                                  |                        |
|                                                          | CFRP                  | -                          |  | Dry installation of fastener except through skin on pressurized areas only<br>Interfay sealant if attachment to the skin on pressurized areas only  | 3                                                                                |                        |
|                                                          | Titanium              | -                          |  | Dry installation of fastener except through skin on pressurized areas only<br>Interfay sealant if attachment to skin on pressurized areas only      | -                                                                                |                        |
|                                                          | CRES                  | Passivation                |  |                                                                                                                                                     | -                                                                                |                        |
| B                                                        | Aluminium             | TSA <sup>1</sup> + P       |  | Wet installation of fastener on tank boundaries<br>Interfay sealant on tank boundaries<br>Fillet sealing on tank boundaries                         | S in tank boundaries<br>for non-self-sealing<br>nuts.<br>P if self-sealing nuts. |                        |
|                                                          | CFRP                  | P                          |  |                                                                                                                                                     |                                                                                  |                        |
|                                                          | Titanium              | P                          |  |                                                                                                                                                     |                                                                                  |                        |
|                                                          | CRES                  | Passivation + P            |  |                                                                                                                                                     |                                                                                  |                        |
| C1                                                       | Aluminium<br>(clad)   | TSA <sup>1</sup><br>P + TC |  | Wet installation of fastener                                                                                                                        | P + TC<br>or V<br>or CI DAP                                                      |                        |
|                                                          | Aluminium<br>(unclad) | TSA<br>P + TC              |  | Interfay sealant if attachment to skin on pressurized areas only                                                                                    |                                                                                  |                        |
|                                                          | CFRP                  | -                          |  | Dry installation of fastener except through skin on pressurized areas only<br>Interfay sealant if attachment to skin on pressurized areas only      | 3                                                                                |                        |
|                                                          | Titanium              | -                          |  | Dry installation of fastener except through skin on pressurized areas only                                                                          | -                                                                                |                        |
|                                                          | CRES                  | Passivation                |  | Interfay sealant if attachment to skin on pressurized areas only                                                                                    | -                                                                                |                        |
| C2-1                                                     | Aluminium             | TSA<br>P + TC              |  | Wet installation of fastener<br>Interfay sealant mandatory<br>Fillet sealing except clip/frame & clip/stringer connection on pressurized areas only | P + TC<br>or P + V<br>or CI DAP                                                  |                        |
|                                                          | CFRP                  | -                          |  | Dry installation of fastener except through skin on pressurized areas only<br>Interfay sealant if attachment to skin on pressurized areas only      | 3                                                                                |                        |
|                                                          | Titanium              | -                          |  | Dry installation of fastener except through skin on pressurized areas only                                                                          | -                                                                                |                        |
|                                                          | CRES                  | Passivation                |  | Interfay sealant if attachment to skin on pressurized areas only<br>Fillet sealing                                                                  | -                                                                                |                        |
| C2-2                                                     | Aluminium             | TSA<br>P + TC              |  | Wet installation of fastener<br>Interfay sealant mandatory<br>Overlap sealing                                                                       | P + TC<br>or P + V<br>or CI DAP                                                  |                        |
|                                                          | CFRP                  | -                          |  | Dry installation of fastener except through skin on pressurized areas only<br>Interfay sealant if attachment to skin on pressurized areas only      | 3                                                                                |                        |
|                                                          | Titanium              | -                          |  | Dry installation of fastener except through skin on pressurized areas only                                                                          | -                                                                                |                        |
|                                                          | CRES                  | Passivation                |  | Interfay sealant if attachment to skin on pressurized areas only<br>Fillet sealing                                                                  | -                                                                                |                        |

Post assembly sealing protection: Elastic Varnish according to Chapter 7.8.

Post assembly sealing protection: Elastic Vamish according to Chapter 7.8.

**Alternatives to Cadmium Plating:**

- **Electrolytic Zinc - Nickel plating**
- **Ion Vapor deposition of aluminum**
- Deposition of zinc by thermal spray process
- Deposition of aluminum by the thermal spray process

**Alternatives to Hard Chromium:**

- Physical Vapor Deposition (PVD) for WC C: H coating
- AMS2454 : Electroless (Chemical) Nickel Phosphorus - PTFE duplex coating



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10. CAN13001: PROCESOS DE PROTECCION SUPERFICIAL
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